

This working paper on the Pandemic Risk Index (PRI) was developed in collaboration with the Bank of England, the Confederation of British Industry (CBI), Experian and partner universities. The project was funded by UK Research and Innovation (UKRI), grant number ES/V015419/1.

UK SMEs: quantifying their pandemic risk and credit risk exposures in the wake of the COVID-19 crisis

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1 Introduction

Covid-19 is bringing devastating challenges to small- and medium-sized enterprises (SMEs) and microenterprises, where SMEs are firms with less than 250 employees and microenterprises are firms with less than 10 employees. A recent statistical insight of the Organization for Economic Cooperation and Development (OECD) reveals that global prolonged coronavirus (COVID-19) lockdown measures put SMEs at particularly high risk of failure. Among all sectors that are directly affected by Covid-19, SMEs account for 75% of all job placements. Strikingly, microenterprises incur the most serious risk of cash shortages. It demonstrates that across OECD countries, SMEs in the UK experience one of the largest effects¹.

¹<https://www.oecd.org/sdd/business-stats/statistical-insights-small-medium-and-vulnerable.htm>.

SMEs have long been recognised as a force contributing to more than one-half of the world's GDP and two-thirds of employment ([Ahmed, Beck, McDaniel and Schropp, 2015](#)), but they faced tremendous credit constraints on securing funding for financing their operations because they lacked sufficient track records of credit and relevant data. Covid-19 reveals the long-term difficulties of SMEs in financing applications. Although the governments in some countries like the UK promptly introduce effective measures to fill SMEs' financing gaps through the schemes of government loan/guarantee, these measures unavoidably carry their inherent shortages. First, such policies are emergent and short-term remedies. The government cannot afford to maintain these policies, especially during the period when the whole economy is under devastating damage due to Covid-19. Second, the risk profiles of SMEs are diverse and it is necessary to evaluate SMEs' credit profiles before issuing them loan due to the requirement of risk control.

Meanwhile, Big Data and Artificial intelligence (AI) are bringing significant changes to our lives. Technological developments in Big Data and AI analysis have the potential to fill the SME financing gap. Nevertheless, there is a lack of an investigation into the effects of the state-of-the-art AI technologies on the financing activities of SMEs and microenterprises. Hence, the objective of this strand is to examine the impact of the AI technologies on SMEs through the lens of the innovative applications of Big Data and AI to SME credit assessment, focusing specifically on the use of Big data and quantitative machine learning (ML) models of default risk to enrich the analysis of SME credit profile.

ML and DL are the fields at the forefront of AI that can be applied to assess an SME's credit profile. ML makes computers learn from company data that serve as an experience and then computers make predictions about

the SMEs' future credit profile. DL further uses a cascade of multiple layers of learning for the extraction of the SME's credit features, where the previous layer's output is the input to its successive layer ([Cagala and Bundesbank, 2017](#)). In short, Covid-19 exaggerates the challenges of financing constraints that are inherent in SMEs and microenterprises. It is a critical task to provide an accurate evaluation of an SME borrower's risk profile. Big Data and AI have the power and potential to fulfil this task.

2 Literature review

2.1 Literature review: PRI

2.1.1 COVID-19 and its economic aftermath

As stressed by [Brodeur, Gray, Islam and Bhuiyan \(2020\)](#), data from National Bureau of Economic Research (NBER) shows that “By the end of May 2020, there had been 106 papers related to COVID-19.” (p. 3 op. cit.). This considers only NBER working papers. For this reason, this literature review is clearly selective.

We start from papers aiming to measuring the impact of COVID-19. These papers use different variables as indicators of the intensity of the COVID-19 crisis:

- Number of COVID-19 related deaths
- Number of COVID-19 related cases
- Case Fatality Rate (CFR): COVID-19 related deaths divided by number of cases

Other papers focus on measuring the economic impact of COVID-19. Rather than using indicators related to how COVID-19 is spreading across different

regions and countries, they focus on the economic measures implemented by governments and local authorities. In particular, social distancing measures can have a profound impact on the economic growth of the regions where “lockdowns” are in place.² However, because potential interfirm relationships related to the supply chain, even regions that are unaffected by lockdowns could suffer. This is true both at the national level and at the international level, because of the global nature of supply chains in the 21st century.

Since economic data is usually available with a lag, in the first half of 2020 some empirical studies concentrated their analysis on the impact of the crisis on the stock market. While some of the results are mixed, there seems to be consensus on two findings:

1. There is a connection between social distancing measures (especially the ones implemented in Italy in February 2020), and stock market returns and volatility (both realized and expected).
2. Economic policies introduced to provide economic stimulus helped the stock market recover.

Measuring the impact of COVID-19

Brodeur et al. (2020) provide a literature review about the economic impact of COVID-19 and is dated June 2020. As of the time of writing, more papers have, of course, been published. This paper, however, constitutes a

²“Lockdowns” usually refer to government measures which force businesses that are not essential for the survival of citizens to shut down. However, different countries employ lockdowns according to different degrees of severity. In the UK, a UK-wide lockdown occurred in March-May 2020, and it was followed by less severe tier-based restrictions. As of January 2021, there are different rules, depending which of the four nations is considered (see <https://www.bbc.co.uk/news/explainers-52530518>).

very important source of information and useful references on the economic impact of COVID-19. A key feature of this paper is a discussion of the different types of datasets used in this type of economic research. Most of the articles rely on publicly available data on the number of confirmed cases and deaths, as well as the number of tests. The ratio of number of confirmed deaths divided by the confirmed number of cases, called the Case Fatality Rate (CFR), can be used to assess the mortality rate of COVID-19, although the number of confirmed cases, of course, depends on the number of tests run, and thus it might be downward biased ([Roser, Ritchie, Ortiz-Ospina and Hasell, 2020](#)). Moreover, the testing capacity was limited during the first stage of the pandemic, meaning that many COVID-19 cases might have been missed. Finally, since the number of tests changes over time asynchronously for different countries, international comparisons might be difficult. [Liang, Tseng, Ho and Wu \(2020\)](#) show that there is a negative correlation between the number of tests and the CFR, confirming that the CFR might be upward biased as a result of a downward-biased number of cases.

Social distancing measures

Given the importance of social distancing measures for economic growth, empirical papers used a variety of indicators linked to government restrictions on social interactions. [Brodeur et al. \(2020\)](#) categorise the proxies used by empirical researchers into three groups: proxies related to the degree of mobility of the population (e.g., Google Mobility reports); modelling proxies related to the spread of the infection; the creation of indices based on the implementation of social-distancing policies. The third type of proxies is particularly useful for the purpose of this project, because the main objective is to form a “pandemic risk index” for SMEs in the UK. For example, [Baker, Bloom,](#)

[Davis and Terry \(2020b\)](#) introduce a COVID-19 index that captures the degree of uncertainty during the COVID-19 crisis. This index is based on several factors: stock market volatility, newspaper-based economic uncertainty, and subjective uncertainty in business expectation surveys.

As reported by [Brodeur et al. \(2020\)](#), social-distancing measures implemented across the world have been quite diverse and, as mentioned above, in the UK different regions have been under different types of restrictions since the COVID-19 outbreak started. Support for these measures varies according to political views, and generally speaking, it appears that right-wing parties tend to be less likely to support stricter restrictions.³

Have these social-distancing measures been effective?

[Jinjarak, Ahmed, Nair-Desai, Xin and Aizenman \(2020\)](#) provide evidence that more stringent pandemic policies were associated with lower mortality growth rates in the first four months of 2020. The success of these policies in curbing mortality rates was stronger for countries with a greater proportion of the elderly population and urban population, greater democratic freedoms, and larger international travel flows. The paper also provides evidence that stricter policies before the first death resulted in lower peak mortality rates and a lower duration to the peak mortality rate. This finding suggests that stricter policies (measured using the Stringency Index, SI)⁴ were beneficial.

Propagation channels

³For example, restrictive measures in the US depends on the party that is in charge in a certain state: states with Democratic governors are 50 percent more likely to implement lockdown/stay-at-home orders ([Baccini and Brodeur, 2020](#)).

⁴<https://www.bsg.ox.ac.uk/research/research-projects/coronavirus-government-response-tracker>

In addition to the stock market, the COVID-19 crisis can affect the economy because it impacts the global supply chain (Bonadio, Huo, Levchenko and Pandalai-Nayar, 2020) . Social distancing measures imposed by governments result in non-working days, affecting economic growth (Mulligan, 2020), but they can be partially offset by measures such as contract-tracing and risk-mitigation measures in the workplace. A contraction in consumption might also occur, although at the beginning of the crisis it seems that expenditure on food and retail increased (e.g., Baker, Farrokhnia, Meyer, Pagel and Yannelis (2020c)).

As mentioned by Inoue and Todo (2020), natural disasters can impose negative economic shocks on regions that are not directly affected by the disaster through the interfirm supply chain channel. Using data for Japan, the authors examine the propagation channel of lockdowns in terms of value-added production. They find that the impact of a lockdown in Tokyo on regions outside of Tokyo that connected to firms in Tokyo via interfirm supply chain linkages would be larger than in Tokyo itself. The authors assume that during the lockdown would affect all sectors apart from essential activities for the survival of citizens.

Impact on the stock market

Given the low frequency of macro-economic data (e.g., GDP and inflation) and financial reports (especially for SMEs), a large part of the literature on the economic impact of COVID-19 focusses on the impact of the COVID-19 on the stock market. Understanding the impact of COVID-19 on the stock market is useful because of two reasons: the stock market incorporates expectations with respect to the impact of COVID-19 on the real economy; stock market crashes and liquidity dry ups on the stock market might have repercussions

on the real economy ([Carlsson-Szlezak, Reeves and Swartz, 2020](#)).

As [Baker, Bloom, Davis, Kost, Sammon and Viratyosin \(2020a\)](#) point out, the restrictions on economic activity and voluntary social distancing since the beginning of the crisis, before the imposition of compulsory lockdown measures, are to blame for the economic impact of the COVID-19 turmoil on the US stock market. [Onali \(2020\)](#) and [Onali and Mascia \(2020\)](#) show that the US stock market experienced a sudden increase in volatility expectations (proxied by the VIX) the week after Italy imposed lockdown measures on Northern Italian towns (Sunday 23 February 2020), confirming the link between lockdown measures and investors' expectations. However, the futures market seems to have under-estimated the potential impact of the crisis in the early stages of the pandemic ([Cheng, 2020](#)) .

Several papers, such as [Ramelli and Wagner \(2020\)](#) and [Ding, Levine, Lin and Xie \(2020\)](#), analyse the factors affecting the impact of COVID-19 on stock prices, focussing on the first quarter of 2020, similar to [Onali \(2020\)](#) and [Onali and Mascia \(2020\)](#). The former paper reports that in the first stage of the crisis, exports and supply-chain exposure to China was the main driver of a negative price reaction to the crisis. As the crisis spread to the US and Europe, factors such as corporate debt and cash holdings became more important. In particular, firms with low levels of cash and high leverage performed worse than firms with high levels of cash and low leverage. The latter paper also find that low cash holdings and high leverage had a negative effect on the price reaction during the first stage of the crisis. They also provide evidence that exposure to the global supply chain was an important driver of the price reaction. Corporate Social Responsibility (CSR) activities and a low level of managerial entrenchment also contributed to softening the impact of the COVID-19 crisis. Notably, the findings related to the impact of CSR and

managerial entrenchment are consistent with those provided by [Albuquerque, Koskinen, Yang and Zhang \(2020\)](#), who find that Environmental and Social (ES) variables matter: stocks with higher ES scores have significantly higher returns, lower return volatility, and higher operating profit margins during the first quarter of 2020.

As said above, [Albuquerque et al. \(2020\)](#) provides evidence even on volatility, although it is not the main focus of the paper. Several empirical papers focus on the impact of the COVID-19 crisis on volatility and other proxies of stock risk. For example, [Onali and Mascia \(2020\)](#) investigate the impact of corporate diversification on idiosyncratic and total volatility and idiosyncratic and total skewness, and find evidence consistent with the view that the lockdown in Northern Italy in late February 2020 increased idiosyncratic and total stock risk for US stocks. [Baek, Mohanty and Glambsky \(2020\)](#), find that the industries that experienced the larger (smaller) increase in risk are petroleum and natural gas, restaurants, hotels and lodgings (food production, beer and liquor).

Economic impact of COVID-19 in the UK

The economic impact of COVID-19 in the UK has been substantial because of the negative effect of lockdowns and other social distancing measures on economic activities. To ameliorate its negative impact on the economy, and especially on SMEs, the UK government announced on 20 March 2020 the Job Retention Scheme (JRS), also known as “furlough” scheme. [Mayhew and Anand \(2020\)](#) argue that the JRS was not enough because of the likelihood that economic recovery would be slow and a rise in unemployment would hit mainly young people. They also stress that social distancing rules are more likely to slow down recovery in industries such as, for example, hospitality,

retail, leisure and entertainment, transport and the Higher Education (HE) sector. Finally, they highlight that without financial help a continuation of social distancing rules will cause companies to lay off workers permanently (instead of temporarily by using the “furlough” scheme), and some firms might also go bankrupt. Mayhew and Anand’s paper was written in May 2020, and thus it does not consider the effect of subsequent policies, in particular the social distancing rules based on Tiers, the economic stimulus provided by the Winter Economy Plan, and the third lockdown started in January 2021.

Cominetti, Gardiner and Slaughter (2020) confirm that the lockdown has had a negative impact in labour-intensive sectors, such as hospitality, and people without permanent jobs, as well as self-employed people. They also confirm that many of these people are young, as outlined by Mayhew and Anand (2020).

In addition to young people, people with temporary contracts, and people working in certain sectors, several contributions argue that the economic crisis during the COVID-19 outbreak has hit self-employed people particularly hard (Blundell, Machin and Ventura, 2020; Cominetti et al., 2020; Reuschke, Henry and Daniel, 2020). Blundell et al. (2020) argue that the Self-employment Income Support Scheme (SEISS) provided critical financial support to self-employed people, but some of them were excluded from the support. Moreover, there is a lack of awareness of other government schemes to tackle the crisis (e.g., the Coronavirus Business Interruption Loan Scheme (CBILS) – see below).

Several schemes targeted SMEs⁵:

- Bounce Back Loan Scheme (BBLs) – This scheme focuses on loans to

⁵For more information: <https://www.gov.uk/government/publications/winter-economy-plan/winter-economy-plan>.

SMEs. Loans are between £2,000 and £50,000, but they cannot be more than 25% of turnover. The key characteristic of this loans is that there is a 100% government guarantee to the lending institution, and the government also pays the first twelve months' interest payments. Under this scheme, businesses can repay their loans over a ten-year period.

- Coronavirus Business Interruption Loan Scheme (CBILS) – This scheme is for SMEs with turnover under £45 million, loans can be up to £5 million, and there is an 80% government guarantee to the lender. Even in this case, the government covers the interest payments for up to a year.

- Future Fund – This scheme is for fast-growing UK-based businesses, and the government loans are subject to matching from private investors.

2.1.2 A review of natural disaster risk indices

Indices can be constructed on linear risk events, non-linear risk events (events that occur with some frequency but have binomial, bimodal or other type of non-stepwise/non-differentiable distributions), or extreme events. This, of course, assumes that these risks are measurable (i.e. is not uncertainty).

Initially, our focus will be on the last type. Pandemics have historically occurred with certain frequency (globally), but potentially the novel component for UK industry is the complete lockdown. Thus, when we speak about pandemic risk index, we wish to separately capture the exposure prior to the policy intervention jointly with the policy intervention (i.e. the effects from the disease/spread plus the lockdown).

[Davidson and Shah \(1997\)](#) construct an earthquake composite risk index (Earthquake Disaster Risk Index) from cities worldwide, which allows for inter-city allocation of mitigation resources and efforts. The index will highlight the

fact that even cities/regions with low seismicity can experience earthquakes, and thus measure how small risk events can have major effects. (Whether an event is a disaster or not can be defined *quantitatively*: How many deaths has been caused or does it require humanitarian assistance? Or *qualitatively*: Event that creates demand upon community system that can't be met by its traditional institutional structure.)

It is an index measuring the risks from an earthquake disaster, not the expected impact of future earthquakes. In other words, it measures factors contributing to earthquake disaster risk (frequency of earthquake, structural vulnerabilities), not the expected consequence from the event (e.g. economic loss, deaths). Therefore, disaster is not only a function of physical impact of an earthquake, but also the capacity of the city to sustain that impact, and the implications of the impact to the city and globally. Also, it is not a loss function estimation.

The factors that have been identified as contributing to a higher disaster risk are:

- Hazard (severity, extent and frequency of the geological trigger phenomenon). This is where the probability is incorporated, which includes the probabilistically expected level of ground shaking for 2 different time periods. The inherent choice of the 2 time periods will define how probable an event is through its return (the larger the time window, the higher the probability). Note that distributional assumptions are not made on the index itself as authors do not take a stance on what constitutes a disaster and what does not.
- Exposure (size of the city, population size, physical objects, amount and type of activities they support)

- Vulnerability (how affected will the people, physical objects or activities be in the short/long-run)
- External context (second-order effects: how impact within a city affects people/activities/objects outside the city)
- Emergency response and recovery capability

The sub-factors contained within each of the factors are:

- Hazard: Ground shaking, collateral hazards
- Exposure: Physical infrastructure, population, economy, social-political systems
- Vulnerability: Physical infrastructure, population, economy, social-political systems
- External context: Economy, transportation, politics, culture
- Emergency response and recovery capability: Planning, resources, mobility and access

For some of these, questionnaires/surveys were conducted to gather technical insights/inputs to better identify indices related to these factors.

Method to construct the index:

1. Create a conceptual framework of all factors that contribute to earthquake disaster risk (geological, engineering, economic, social, political, cultural)
2. Identify simple measurable indicators to represent each factor (e.g. population, GDP/capita, percentage of urbanized area that has soft soil). See p. 68 [Davidson and Shah \(1997\)](#) for an example.

3. Combine the indices mathematically using simple methods. Benefits (or drawbacks) of linear versus non-linear methods are discussed, and multiple approaches to weighting of indicators are being considered.
4. Collect data and validate EDRI factors using sample of cities worldwide. EDRI cannot be directly validated as it is an approximation of the true disaster risk.
5. Sensitivity analysis
6. Interpretation of the EDRI coefficients in a scientifically intuitive manner

The added benefit of the framework is that one can measure the weight (i.e. the contribution) of each factor, across cities, to the overall risk index. It is not dynamic/time-varying. The data requirement is low, but it requires a lot of assumptions on the measured impact, representativeness of indicator to factor, and aggregation. Note also that this approach takes a view that societal or human activity contributes to higher probability of disaster (risk), and so disaster is not only determined by exogenous/physical factors.

2.1.3 Applications of principal component analysis in risk measurement

Principal component analysis (PCA) is a widely used data analysis technique that is specifically designed to simplify working with high dimensional data. By identifying the main orthogonal factors that explain the majority of the variation in a data set, PCA helps to reduce dimensionality of large data sets, hence making them easier to analyse. Mathematically, the eigenvectors with the largest eigenvalues of the covariance matrix of a certain set of variables are equal to the data set's principal components.

Within the finance and risk literature, several studies have used PCA to estimate portfolio risk and systemic risk. For instance, [Billio, Getmansky, Lo and Pelizzon \(2012\)](#) construct an interconnectedness measure for the financial system, named the “Cumulative Risk Fraction”, based on the covariance matrix of the financial institutions’ stock returns. Whenever a lower number of PCs are required to explain a certain level of interconnectedness within the system, the systemic risk is deemed to be higher. Other studies such as [Kritzman, Li, Page and Rigobon \(2011\)](#), who introduce the absorption ratio, and [Pasini \(2017\)](#) use very similar methodologies to examine systemic and portfolio risk. In a similar manner, [Rodríguez-Moreno and Peña \(2013\)](#) apply PCA to a portfolio of bank Credit Default Swap (CDS) spreads and conclude that the first PC performs the best as a system-wide measure of systemic risk.

Another approach within the applications of PCA in risk assessment (in addition to portfolio analysis), is the use of PCA to build indices based on a set of explanatory variables. [Dai, Xiong and Zhou \(2021\)](#) construct a global economic policy uncertainty index via PCA of the uncertainty indices of the world’s major economies. [Chinn and Ito \(2008\)](#) use PCA to build a financial openness index to measure the freedom of cross-border financial transactions in a given country. [León and Murcia \(2013\)](#) and [Morales and Estrada \(2010\)](#) employ PCA to derive indices of financial system stability in Columbia using measures of size, connectedness and substitutability ([León and Murcia, 2013](#)) and profitability, liquidity and probability of default ([Morales and Estrada, 2010](#)).

Financial Condition Indices (FCI), also known as Financial Stress Indices (FSI) are important metrics provided by financial authorities and institutions to monitor the financial health of the economy. Among various methodologies, PCA is a common approach to build FCIs and FSIs. The St. Louis Fed’s FSI

(STLFISI) and the Kansas City Fed’s FSI (KCFISI) are two important examples of indices constructed based on PCA (see: [of St Louis \(2020\)](#); [Amol \(2010\)](#); [Hakkio, Keeton et al. \(2009\)](#); [Kliesen, Smith et al. \(2010\)](#)). The STLFISI is estimated using the first PC of 18 variables that are categorised as interest-rate-related variables, yield spreads (for various fixed income securities) and other volatility- and market-related variables. The KCFISI is constructed in a similar manner using 11 variables related to yield spreads and the behaviour of asset prices. The Deutsche Bank FCI is another financial stability metric that is constructed via PCA using exchange rate, bonds, stocks and housing market indicators ([Hooper, Mayer and Slok, 2007](#); [Hooper, Slok and Dobridge, 2010](#)).

Lastly, [Hatzius, Hooper, Mishkin, Schoenholtz and Watson \(2010\)](#) employ PCA to build a comprehensive FCI that is able to include a relatively larger number of variables for a longer period of time, thanks to techniques that facilitate working with unbalanced panels. Moreover, their approach allows for disentangling endogenous macroeconomic trends and policy influences from pure financial shocks, hence further improving the index’s predictive power with respect to financial shocks. The comparisons in the study show that this index has a higher predictive power than the simpler counterparts. Other studies such as [Darracq Pariès, Maurin and Moccero \(2014\)](#) also follow the [Hatzius et al. \(2010\)](#) methodology to construct FCIs.

2.2 Literature review: Machine/deep learning methods in credit assessment

Among various sources of risk to which financial institution is exposed, one of the most challenging risk is credit risk. For credit risk evaluation, the main

analytical technique is credit scoring. An increasing number of credit scoring models with better performance is achieved by the application of artificial intelligence.

[Ghodselahi and Amirmadhi \(2011\)](#) design a hybrid model for credit scoring and the decisions of credit granting by applying ensemble learning. They utilize ten classifier estimators as the members to construct their ensemble model. To begin with, they compare the accuracy in classification of three base classifiers: Support vector machine, Neural Networks and Decision Tree. They find that even a slight improvement in the accuracy of the credit score leads to a significant reduction of loss, which implies that it is of great importance to utilize the best classification model.

[Ghodselahi and Amirmadhi \(2011\)](#) use a German dataset that contains 1000 instances of credit application cases, where 700 cases were granted and 300 cases were rejected. The dataset characterizes each case by 20 decision attributes including 13 categorical attributes and 7 numerical attributes. Using this dataset, they demonstrate the test of their model and classifiers. Their test results reveal that their proposed hybrid ensemble model achieves a higher classification accuracy and performance than other credit scoring models and methods. They further point out among the three classifiers, the best performance and accuracy is obtained by using the Support Vector Machine.

A basic hypothesis in machine learning is that one can preliminarily restrict the space of high-dimensional input features to only the space with the important features. Based on this hypothesis, [Oreski and Oreski \(2014\)](#) propose a new advanced heuristic algorithm, specifically the neural networks (HGA-NN) based hybrid genetic algorithm. In their preliminary restriction, they employ fast algorithms for running earlier experience and ranking feature. In addition, during the progress of creating the initial population, they made enhancements

by adding an incremental stage in their genetic algorithm.

Using the novel algorithm to identify an optimum feature subset in credit risk assessment, [Oreski and Oreski \(2014\)](#) increase the scalability and classification accuracy. They evaluate the performances of their new HGA-NN classifier by applying it to a Croatian bank’s real-world credit dataset and then further validate their findings on a German credit dataset. The comparison between the classification accuracy of their novel algorithm and that reported in the prior literature shows that in the application of retail credit risk assessment, their new HGA-NN classifier demonstrates a higher performance in the feature selection and classification.

In automatic credit scoring systems, Multi-layer perceptron (MLP) neural networks are widely used to obtain high accuracy and efficiency. In the framework of MLP neural networks, [Zhao, Xu, Kang, Kabir, Liu and Wasinger \(2015\)](#) develop a higher accuracy credit scoring model and train the model with the back propagation algorithm. Their method achieve three improvement in the credit scoring models. First, they use a new method of Average Random Choosing to optimize the data distribution in datasets. Second, they compare the effects produced by different training-validation-test instance numbers. Third, they identify the most suitable number of hidden units. Using different initial weights and training instances, they trained 34 models 20 times, where each model has 6 to 39 hidden units with one hidden layer.

Using a German credit dataset, [Zhao et al. \(2015\)](#) show that their model has a classification accuracy of 87% and is higher by 5% than the best result recorded in the previous literature. One of the reason for this improvement is that compared with traditional methods, their optimization of dataset structure is helpful in enhancing a model’s accuracy significantly. Furthermore, they illustrate the tendency in their models’ scoring accuracy when they in-

crease the number of hidden units. They show that their novel method can be applied to imbalanced dataset and many MLP neural network applications including credit scoring.

The survey article [Dastile, Celik and Potsane \(2020\)](#) uses a systematic literature survey approach to give a systematical review on the applications of statistical models and machine learning models in credit scoring. The survey builds a guiding framework of machine learning and then identifies the limitations in literature and the emerging directions. This literature survey reviews 74 primary studies in journal and conference articles published between 2010 and 2018. Using the meta-analysis, they found that an ensemble of classifiers achieve a better performance than single classifiers. Deep learning models have not been applied widely in credit scoring but they show promising results. Similarly, the aim of the article [Carvalho, Pereira and Cardoso \(2019\)](#) is to present a review of the current research state on machine learning interpretability. It also emphasises the societal impact, the method development, and metrics of measures. It finally points out the future directions of work on this field.

[Hashemi and Fathi \(2020\)](#) study the existing operational methods in ML in the extrapolation of a retail customer’s default probability. They use PermuteAttack to predict the default probability of a loan based on customer information. They apply the approach to German credit dataset that contains 20 features with 13 categorical ones including sex, loan purpose, and housing type. A scikit-learn RandomForest model with 10 decision trees are used. The categorical features are one-hot encoded, creating 61 features in total. There are 1000 samples in total, which are split into 60% train and 40% test samples. They then draw a counterfactual example for a defaulting and non-defaulting customer.

[de Paula, Artes, Ayres and Minardi \(2019\)](#) use a combination of credit scor-

ing and profit scoring to increase the effectiveness of the loan-granting process in credit unions based on the data of personal loans transactions of a Brazilian credit union. They find that statistical methods improves significantly default predictability compared to subjective techniques. The random forests model is superior in estimating credit scoring and profit scoring, compared to logit and ordinary least squares method (OLS) regression.

[Babaev, Savchenko, Tuzhilin and Umerenkov \(2019\)](#) present a new method to credit scoring of retail customers in the banking industry by using on deep learning methods. They apply RNNs to fine grained transnational data to compute credit scores for loan applicants. They show that their method significantly outperforms the baselines based on the customer data of Sberbank. They also carried out a pilot study on loan applicants of the bank, which leads to significant financial gains for the organization.

Ensemble methods based on decision trees, e.g., the random forest method, achieve superior classification performance than standard logistic regression models. Nevertheless, the credit risk industry mainly uses logistic regression as the benchmark because its interpretability meets the requirements of financial regulators well. [Dumitrescu, Hue, Hurlin and Tokpavi \(2018\)](#) achieve the advantages of both methods by introducing a high-performance and interpretable credit scoring method called penalised logistic tree regression (PLTR). Such method improves the performance of logistic regression by using information from decision trees. They use the rules extracted from different short-depth decision trees built with pairs of predictive variables as predictors in the model with penalised logistic regression. PLTR captures non-linear effects in credit scoring data and meanwhile preserves the intrinsic interpretability of the logistic regression model. Their results demonstrate that PLTR obtain significantly more accurate estimates of credit risk than logistic regression and the random

forest method.

3 Data and variables

Our main data source is the FAME database (hereafter, FAME) provided by Bureau van Dijk (a Moody’s Analytics Company). FAME is a part of the Amadeus (European firms) and Orbis (Global firms) databases that Bureau van Dijk provides.⁶ FAME includes data on more than 14 million firms operating in the UK. FAME includes non-financial data (for example, listing status, SME indicator, subsidiaries, ownership structures, and locations of trading addresses) and financial data (for example, balance sheet, income statement, and cash flow statement). The FAME data is sourced from the filings submitted by firms to the Companies House, the official UK firm registrar. Hence, FAME includes data on all legal forms of UK firms (i.e. publicly listed, privately held, limited liability firms, and partnerships). The vast majority of firms in FAME are SMEs. Hence, FAME fits the purposes of this project.

Table 1 presents our sample selection criteria. Our final sample includes firms with trading addresses in England, Wales, Scotland, and Northern Ireland, with available accounts, total assets, and turnover data for the years 2019, 2020 and 2021. Our final sample is composed of 218,858 firms (only 33,977 firms with both pre-covid and post-covid data).⁷

⁶For a more detailed description of the FAME database, see [Joseph, Kneer, Van Horen and Saleheen \(2020\)](#) and [Bahaj, Foulis and Pinter \(2020\)](#).

⁷Our FAME data has been extracted on 5th March 2021.

Table 1. Sample Selection Criteria

Sample Selection Criteria	Number of Firms
All firms in FAME	14,790,141
Firms with primary trading address in England, Wales, Scotland, and Northern Ireland	13,528,551
Firms with available accounts in 2019, 2020 & 2021	3,836,977
Firms with available data on total assets	3,685,595
Firms with available data on turnover	280,107
Final Sample	218,858

This table shows the search criteria used to select our final sample.

Table 2. Variable Definitions

VARIABLES	Definition	Source
A) Firm-Level Financial Variables (measured before 1st April 2020)		
Total Assets	Fixed assets + Current assets. Fixed assets = Tangible assets + Intangible assets + Investments (fixed assets). Current assets = Stock & work-in-progress (WIP) + Trade debtors + Bank & deposits + Other current assets + Investments (current assets). Measurement units: thousand GBPs.	FAME
Fixed Assets%	Fixed assets / Total assets. Measurement units: decimals.	FAME
Tangible Assets%	Tangible assets / Total assets. Measurement units: decimals.	FAME
Debtors%	Trade debtors / Total assets. Measurement units: decimals.	FAME

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
Current Liabilities%	Current liabilities / Total assets. Current liabilities = Trade creditors + Short-term loans & overdrafts + Total other current liabilities. Long-term liabilities = Long-term debt + Total other long-term liabilities + Provisions for other liabilities + Pension liabilities + Balance sheet minorities. Measurement units: decimals.	FAME
Creditors%	Trade creditors / Total assets. Measurement units: decimals.	FAME
Balance Sheet Minorities%	Balance sheet minorities / Total assets. Measurement units: decimals.	FAME
Shareholders Funds%	Shareholders' funds / Total assets. Shareholders' funds = Issued capital + Total reserves. Measurement units: decimals.	FAME
Total Reserves%	Total reserves / Total assets. Measurement units: decimals.	FAME

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
Turnover	Total turnover (Gross revenues/sales). Measurement units: thousand GBPs.	FAME
Overseas Turnover%	Overseas turnover / Total turnover. Measurement units: decimals.	FAME
EBIT%	Operating profit / Total turnover. Operating profit = Gross profit - Administration expenses + Other operating income pre-operating profit + Exceptional items pre-operating profit. Gross profit = Turnover - Cost of sales + Exceptional items pre-gross profit + Other income pre-gross profit. Measurement units: decimals.	FAME
Administration Expenses%	Administration Expenses / Total turnover. Measurement units: decimals.	FAME
Exceptional Items%	Exceptional items pre-operating profit / Total turnover. Measurement units: decimals.	FAME

Table 2: Variable Definitions (continued)

VARIABLES		Definition	Source
ROA		Pre-tax income / Total assets. Pre-tax income = Operating Profit + Total other income & interest received + Exceptional items - Interest paid	FAME
Liquidity Ratio		(Current assets - Stock & WIP) / Current liabilities. Measurement units: decimals.	FAME
Assets Turnover		Total assets / Total turnover. Measurement units: decimals.	FAME
Gearing		(Short-term loans & overdrafts + Long-term liabilities) / Shareholders' funds. Measurement units: decimals.	FAME
Debtors	Collection Days	(Trade debtors / Total turnover) * 365	FAME
Creditors	Payment Days	(Trade creditors / Total turnover) * 365	FAME
Firm-Level Index	Financial	First principal component of the firm-level financial variables.	PCA

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
B) Firm-Level Non-Financial Variables (measured before 1st April 2020)		
SME	1 if the firm is a small or medium-sized enterprise (SME); 0 otherwise.	FAME
Listed Firm	1 if the firm is publicly listed; 0 otherwise	FAME
Firm Age	The number of days since the firm's incorporation	FAME
Number of Employees	Number of employees	FAME
Firm Independence	BvD Independence Indicator (the higher, the more independent the firm is): 4) the firm has no identified shareholder with over 25% ownership, 3) the firm has no identified shareholder with over 50%, but at least one with over 25%, 2) the firm has an identified shareholder with over 50% total ownership or calculated total ownership, 1) the firm has an identified shareholder with over 50% direct ownership or is a branch/foreign company.	FAME

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
Business Diversification	The number of firm's sectors (i.e. two-digit SIC codes)	FAME
Geographical Diversification	The number of firm's subsidiaries and trading addresses	FAME
Firm-Level Financial Index	First Principal Component of the firm-level non-financial variables.	PCA
Firm-Level Overall Index	First Principal Component of the firm-level financial and non-financial variables.	PCA
C) Regional Variables (measured at the NUTS1 region)		
Regional Covid Cases	Cumulative number of new Covid cases from March 2020 until the end of the firm's financial year.	gov.uk
Severe Anxiety	Proportion of people who suffer from medium to high Covid-related anxiety.	ONS
Covid Information	Proportion of people who feel they receive enough Covid-related information.	ONS

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
Self-isolation	Proportion of people who have to self-isolate due to Covid.	ONS
Affordability	Proportion of people who can afford unexpected but necessary expense of £850 after Covid.	ONS
Pessimism	Proportion of people who are either uncertain about or reject the notion that life will return to full normality after Covid.	ONS
Regional Covid Social Index	First Principal Component of the regional Covid-related social variables.	PCA
Regional GDHI Per Head	Gross Domestic Household Income per head (2018).	ONS
D) Sectoral Variables		
Sectoral Covid Exposure Index	Ranking of sectors (two-digit SIC codes) according to Market-Adjusted Returns on 16th & 17th March 2020 (the higher, the more negative sectoral reaction to lockdown news).	Capital IQ

Table 2: Variable Definitions (continued)

VARIABLES	Definition	Source
E) Country-Level Variables		
Country-Level Stringency Index	The Oxford Coronavirus Government Response Tracker (OxCGRT) project calculates a Stringency Index, a composite measure of nine of the response metrics. The nine metrics used to calculate the Stringency Index are: school closures; workplace closures; cancellation of public events; restrictions on public gatherings; closures of public transport; stay-at-home requirements; public information campaigns; restrictions on internal movements; and international travel controls. It is measured from March 2020 until the end of the firm's financial year.	ourworldindata.org

Table 2: Variable Definitions (continued)

VARIABLES		Definition	Source
F) Other Variables			
Pre-Covid Score	Credit	Firm's credit score before 1st April 2020. It predicts the likelihood of a company going insolvent in the next 12 months. It ranges from 0 to 100 with higher scores indicating better solvency.	FAME
Post-Covid Score	Credit	Firm's credit score on or after 1st April 2020. It predicts the likelihood of a company going insolvent in the next 12 months. It ranges from 0 to 100 with higher scores indicating better solvency.	FAME
Post-Covid CCJs		Natural logarithm of the total value of County Court Judgements (CCJs) issued against the company after on or 1st April 2020.	FAME
Post-Covid Unsatis- fied CCJs	Unsatis-	Natural logarithm of the value of County Court Judgements (CCJs) issued against but not paid by the company on or after 1st April 2020.	FAME

4 Research Methodology: PRI

4.1 Principal Component Analysis

4.1.1 Firm-Level Financial Index

To construct the Firm-Level Financial Index (FLFI), we start our PCA with 40 financial variables from the balance sheet and income statement. Due to many missing observations (more than 75% of the final sample), we exclude all variables from the cash flow statement. Apart from total assets (the main variable in the balance sheet) and turnover (the main variable in the income statement), all other variables are presented as ratios. All variables are measured using pre-covid data (i.e. for the accounts years ending before 1st April 2020).

In a pilot PCA, we exclude 20 financial variables with Kaiser-Meyer-Olkin (KMO) statistics of less than 0.40.⁸ Hence, we end up with 20 firm-level financial variables. Variable definitions are presented in Table 2 and summary statistics are presented in Table 3. FLFI has been constructed using 142,527 firms.

Table 4 shows the PCA outcomes for FLFI. Panel A presents the correlation coefficients of the financial variables. Most of the variables are weakly to mediumly correlated, which casts doubt on the validity of PCA as a tool to construct FLFI. However, Panel B shows that KMO statistics for all variables, except for Total Assets, exceed 0.50.⁹ Panel C shows that six principal components have eigenvalues exceeding 1 and explain 59% of the variations in

⁸The list of the 20 financial variables excluded from PCA is presented in Appendix 1.

⁹According to [Kaiser \(1974\)](#), the following labels are given to KMO Statistics: 1) unacceptable (0.00 - 0.49), 2) miserable (0.50 - 0.59), 3) mediocre (0.60 - 0.69), 4) middling (0.70 - 0.79), 5) meritorious (0.80 - 0.89), 6) marvellous (0.90 - 1.00).

Table 3. Summary Statistics

VARIABLES	Obs	Mean	Std. Dev.	P5	P50	P95
A) Firm-Level Financial Variables						
Total Assets	214543	27146.71	114500.2	.5	406.549	101338
Fixed Assets%	214543	.295	.358	0	.089	.98
Tangible Assets%	214543	.189	.302	0	.013	.922
Debtors%	214543	.085	.17	0	0	.481
Current Liabilities%	214543	.701	1.968	0	.277	1.732
Creditors%	214543	.056	.129	0	0	.319
Balance Sheet Minorities%	214543	0	0	0	0	0
Shareholders Funds%	214543	.445	.442	-.243	.482	1
Total Reserves%	214543	-.161	3.375	-1.786	.412	1
Turnover	201177	15442.1	53950.97	2.123	400.832	67313.45
Overseas Turnover%	201458	.036	.152	0	0	.225
EBIT%	201177	-.118	1.624	-.979	.051	.926
Administration Expenses%	201177	.751	1.44	0	.483	1.614
Exceptional Items%	201177	0	.001	0	0	0
ROA	202128	.326	2.99	-.909	.046	1.954
Liquidity Ratio	185513	8.229	26.201	.065	1.497	33.627
Assets Turnover	200551	8.227	33.852	.043	.758	28.54
Gearing	175049	13.345	93.844	0	.02	13.212
Debtors Collection Days	200552	26.389	59.259	0	0	110.518
Creditors Payment Days	200731	18.839	54.475	0	.006	74.477
Firm-Level Financial Index	142527	0	1.773	-2.399	-.173	3.045
B) Firm-Level Non-Financial Variables						
SME	218858	.878	.327	0	1	1
Listed Firm	218858	.006	.079	0	0	0
Firm Age	216636	5645.449	6132.64	545	3707	17112
Number of Employees	137693	208.979	3376.935	1	11	481
Firm Independence	218842	2.152	1.367	1	1	4
Business Diversification	208474	1.201	.543	1	1	2
Geographical Diversification	218858	2.847	28.422	0	1	6
Firm-Level Non-Financial Index	133123	0	1.302	-.63	-.312	1.28
Firm-Level Overall Index	97932	0	1.917	-2.75	-.127	3.219
C) Regional Variables						
Regional Covid Cases	33965	605.914	795.303	143.1	376.3	2175.6
Severe Anxiety	215268	.585	.015	.56	.58	.6
Covid Information	215268	.84	.025	.81	.85	.88
Self-isolation	215268	.272	.017	.25	.27	.29
Affordability	215268	.584	.026	.55	.58	.62
Pessimism	215268	.201	.019	.17	.2	.22
Regional Covid Social Index	215268	0	1.874	-2.448	-.113	2.015
Regional GDHI Per Head	218768	23118.61	4612.091	17340	22205	29362
D) Sectoral Variables						
Sectoral Covid Exposure Index	190580	30.622	17.535	6	24	63
E) Country-Level Variables						
Country-Level Stringency Index	38747	74.631	3.526	70.368	75.311	79.594
F) Other Variables						
Pre-Covid Credit Score	196638	59.071	27.874	22	55	99
Post-Covid Credit Score	184031	55.313	27.957	21	45	98
Post-Covid CCJs	11192	1987.083	19534.39	0	0	4514
Post-Covid Unsatisfied CCJs	11192	1792.714	18344.34	0	0	3595

This table shows the summary statistics of our variables. Variable definitions are presented in Table 2.

the original variables. The first principal component (Comp1) has an eigenvalue of 3.14 and explains 16% of variations in the original variables. Panel D shows the loadings of the variables in the six principal components with the highest eigenvalues. Except for the fourth principal component (Comp4), all principal components have four variables with loadings exceeding 0.30. The main variables loaded in Comp1 are *Shareholders Funds%* (-0.3967), *Current Liabilities%* (0.3857), *Creditors%* (0.3647), and *Debtors%* (0.3615). Our FLFI is constructed using Comp1.

4.1.2 Firm-Level Non-Financial Index

To construct the Firm-Level Non-Financial Index (FLNI), we utilize 7 non-financial variables, which are: 1) SME indicator, 2) publicly listed firm indicator, 3) firm age, 4) number of employees, 5) firm independence (i.e. ownership structure), 6) business diversification, and 7) geographical diversification. All variables are measured using pre-covid data (i.e. for the accounts years ending before 1st April 2020). Variable definitions are presented in Table 2 and summary statistics are presented in Table 3. FLNI has been constructed using 133,123 firms.

Table 5 shows the PCA outcomes for FLNI. Panel A presents the correlation coefficients of the non-financial variables. Most of the variables are weakly to mediumly correlated, which casts doubt on the validity of PCA as a tool to construct FLNI. However, Panel B shows that KMO statistics for all variables, except for Firm Independence, exceed 0.50. Panel C shows that three principal components have eigenvalues exceeding 1 and explain 56% of the variations in the original variables. The first principal component (Comp1) has an eigenvalue of 1.69 and explains 24% of variations in the original variables. Panel

D shows the loadings of the variables in the three principal components with the highest eigenvalues. Except for the second principal component (Comp2), all principal components have four variables with loadings exceeding 0.30. The main variables loaded in Comp1 are *Geographical Diversification* (0.5473), *Number of Employees* (0.5057), *Listed Firm* (0.4285), and *SME* (-0.3787). Our FLNI is constructed using Comp1.

4.1.3 Firm-Level Overall Index

To construct the Firm-Level Overall Index (FLOI), we utilize the 20 financial variables used in constructing FLFI and the 7 non-financial variables used in constructing FLNI. All variables are measured using pre-covid data (i.e. for the accounts years ending before 1st April 2020). Variable definitions are presented in Table 2 and summary statistics are presented in Table 3. FLOI has been constructed using 97,932 firms.

Table 6 shows the PCA outcomes for FLOI. Panel A presents the correlation coefficients of the financial and non-financial variables. Most of the variables are weakly to mediumly correlated, which casts doubt on the validity of PCA as a tool to construct FLOI. However, Panel B shows that KMO statistics for all variables exceed 0.50 with many variables having KMOs that exceed 0.70. Panel C shows that eight principal components have eigenvalues exceeding 1 and explain 58% of the variations in the original variables. The first principal component (Comp1) has an eigenvalue of 3.67 and explains 14% of variations in the original variables. Panel D shows the loadings of the variables in the eight principal components with the highest eigenvalues. Except for the fifth principal component (Comp5), the seventh (Comp7) and eighth (Comp8), all principal components have four variables with loadings exceeding

0.30. The main variables loaded in Comp1 are *Shareholders Funds%* (-0.3552), *Current Liabilities%* (0.3544), *Creditors%* (0.3255), and *Debtors%* (0.3153). Our FLOI is constructed using Comp1.

4.1.4 Regional Covid Social Index

To construct the Regional Covid Social Index (RCSI), we start our PCA with 14 variables extracted from the surveys conducted by the Office for National Statistics (ONS) about the Coronavirus and the social impacts on households in subnational areas in Great Britain.¹⁰ There have been three rounds of data collection for this survey: 1) 20th March to 31st May 2020, 2) 4th June to 2nd August 2020, and 3) 7th January to 28th March 2021. The collective results of this survey have been published on 27th April 2021 and used to construct our RCSI. The survey data is available at the NUTS1 region level¹¹ (except for Northern Ireland, which has no data in the survey¹²).

In a pilot PCA, we exclude 9 social variables with Kaiser-Meyer-Olkin (KMO) statistics of less than 0.40.¹³ Hence, we end up with 5 social variables. Variable definitions are presented in Table 2 and summary statistics are presented in Table 3. RCSI has been constructed using 215,268 firms.

Table 7 shows the PCA outcomes for RCSI. Panel A presents the correlation coefficients of the social variables. Most of the variables are mediumly to

¹⁰Data for this ONS survey can be found here:

<https://www.ons.gov.uk/releases/...britain2020and2021>.

¹¹The UK has 12 NUTS1 regions, which are: Wales, Scotland, Northern Ireland, and the nine NUTS1 regions of England (North East, North West, Yorkshire and the Humber, East Midlands, West Midlands, East of England, London, South East, and South West).

¹²Our final sample includes only 3,500 firms in Northern Ireland. This accounts for 1.5% of the final sample size.

¹³The list of the 9 social variables excluded from PCA is presented in Appendix 2.

strongly correlated, which supports the validity of PCA as a tool to construct RCSI. Moreover, Panel B shows that KMO statistics for all variables exceed 0.60. Panel C shows that only the first principal component has an eigenvalue exceeding 1 (that is, 3.51) and explains 70% of variations in the original variables. Panel D shows the loadings of the variables in the first principal component (Comp1). All five variables have loadings exceeding 0.30. Our RCSI is constructed using Comp1.

Noteworthy is that adverse social variables (*Severe Anxiety*, *Self-isolation*, and *Pessimism*) are loaded positively, while favourable social variables (*Covid Information* and *Affordability*) are loaded negatively. Collectively, RCSI can be viewed as a composite measure of the adverse social impact of Covid on households in the UK.

4.2 Regional Covid Cases

To account for the proximity of the firm to covid hot spots, we use official data on cumulative number of covid cases at the NUTS1 region level.¹⁴ Variable definition is presented in Table 2 and summary statistics are presented in Table 3. For each firm, the variable *Regional Covid Cases* is measured at the end of the respective accounts year (for example, if a firm has its primary trading address in London and its accounts year ends on 31st December 2020, *Regional Covid Cases* reflects the cumulative number of covid cases in the London region from the earliest date covid data is available until 31st December 2020).

¹⁴Data on new covid cases can be found here: <https://coronavirus.data.gov.uk/details/cases>.

4.3 Regional GDHI Per Head

To account for differences in household finance across UK regions, we control for the regional gross disposable household income (GDHI), which is defined as “*The amount of money that all of the individuals in the household sector have available for spending or saving after income distribution measures (for example, taxes, social contributions and benefits) have taken effect.*”(ons.gov.uk, 2021). Variable definition is presented in Table 2 and summary statistics are presented in Table 3. The variable *Regional GDHI Per Head* is measured in 2018 (the most recent data available on GDHI in the UK).

4.4 Sectoral Covid Exposure Index

To construct the Sectoral Covid Exposure Index (SCEI), we look at the average market-adjusted stock returns (MARs) at the two-digit SIC level around the announcement of the first UK national lockdown on 23rd March 2020. Stock returns are collected from the Capital IQ database. As illustrated in Figure 1, the largest drops in the UK stock markets occurred on 16th March (-4.46%) and 17th March (-7.09%). The drop was most likely due to UK Prime Minister’s announcement of restrictions to curb spreading of Covid. The UK PM is quoted “*now is the time for everyone to stop non-essential contact with others and to stop all unnecessary travel.*”¹⁵ Therefore, we use MARs(-5,-4) (i.e. cumulative market-adjusted returns on 16th and 17th March 2020) to rank the sector-level covid exposure in the UK.

¹⁵The PM speech can be found here: <https://www.gov.uk/government/speeches/pm-statement-on-coronavirus-16-march-2020>.

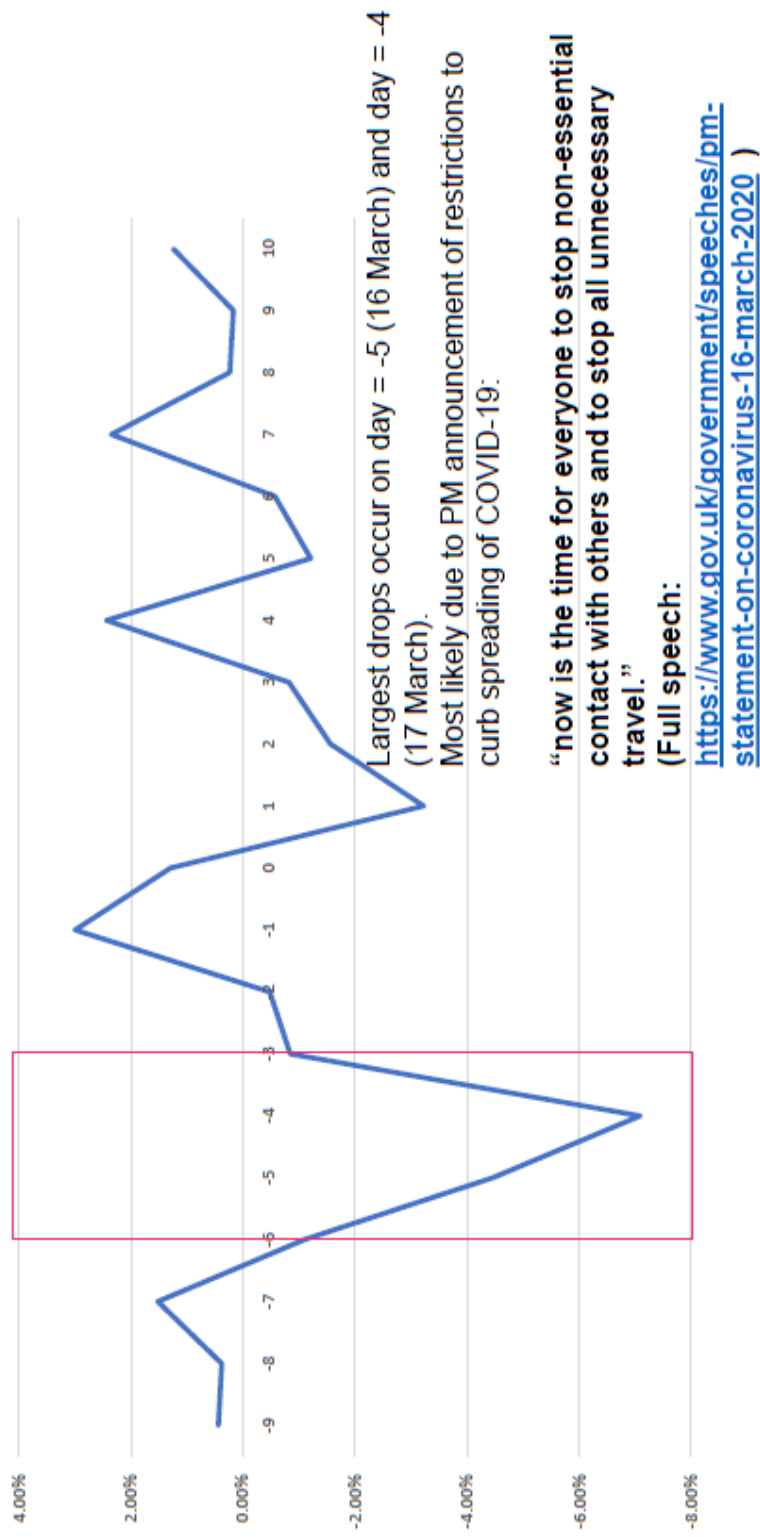


Figure 1. Stock Market Reaction to the First UK National Lockdown

This figure illustrates the average market-adjusted stock returns of UK firms around the first UK national lockdown announcement on 23rd March 2020.

Table 8 shows the ranking scores of SCEI. The "Local and interurban passenger transit" sector (SIC: 41) was the most adversely impacted with an overall plunge of 46.09%. The "Insurance agents, brokers, & service" sector (SIC: 64) was the least adversely impacted with an overall surge of 1.89%. Overall, 62 out of 66 sectors have experienced negative MARs(-5,-4) with only four sectors (Tobacco products, General merchandise stores, Insurance agents, brokers, & service, and Non-classified sectors) having experienced slightly positive MARs(-5,-4). Collectively, SCEI can be viewed as a composite measure of the adverse economic impact of Covid on business sectors in the UK.

4.5 Country-Level Stringency Index

To control for covid-related restrictions at the country level, we include in our empirical models the Country-Level Stringency Index (CLSI) extracted from the Oxford COVID-19 Government Response Tracker (OxCGRT) developed by [Hale, Angrist, Goldszmidt, Kira, Petherick, Phillips, Webster, Cameron-Blake, Hallas, Majumdar et al. \(2021\)](#). OxCGRT captures variations in governmental responses and policies towards covid using 9 metrics, which are: 1) school closures; 2) workplace closures; 3) cancellation of public events; 4) restrictions on public gatherings; 5) closures of public transport; 6) stay-at-home requirements; 7) public information campaigns; 8) restrictions on internal movements; and 9) international travel controls. It takes values from 0 to 100 with higher scores indicating more stringent covid restrictions at the country level. The variable *Country-Level Stringency Index* captures OxCGRT scores for the UK. For each firm, it is measured as an average of the daily OxCGRT (UK) scores from 23rd March 2020 to the end of the respective accounts year.

4.6 Other Variables

4.6.1 Credit Scores

The firm’s credit score predicts the likelihood of a company going insolvent in the next 12 months. It ranges from 0 to 100 with higher scores indicating better solvency. Data on credit scores are collected from FAME. The credit scores available on FAME have been developed by Jordans Business Information Services in collaboration with CRIF Decision Solutions Ltd. The primary sources of data to compute the credit scores are company accounts, county court judgements, directors, and shareholders. In our empirical models, we include *Post-Covid Credit Score* (measured on or after 1st April 2020) as a dependent variable and *Pre-Covid Credit Score* (measured before 1st April 2020) as an independent variable. Variable definitions are presented in Table 2 and summary statistics are presented in Table 3.

4.6.2 County Court Judgements

The firm’s county court judgement (CCJ) is a type of court order that means the firm formally owes a certain amount of money that has to be paid by a specific deadline. Unless the CCJ is paid in full within 30 days of receiving the judgment, it becomes an unsatisfied CCJ and is entered on the firm’s credit record at the Register of Judgments, Orders and Fines, and it remains there for six years. Both CCJs and unsatisfied CCJs are strong indicators of unstable financial condition and can seriously affect the ability of the firm to secure external finance. Therefore, we include *Post-Covid CCJs* and *Post-Covid Unsatisfied CCJs* as dependent variables in our empirical models. Both variables are measured on or after 1st April 2020. Since both variables are highly skewed, we use their log-transformations. Variable definitions are

presented in Table 2 and summary statistics are presented in Table 3.

5 Research Methodology: Machine/deep learning methods in credit assessment

We apply ML and DL to assess an SME’s credit profile. ML makes computers learn from company data and then computers make predictions about the SMEs’ future credit profile. DL further uses a cascade of multiple layers of learning for the extraction of the SME’s credit feature. Specifically, an application of ML to SMEs’ credit assessment comprises five steps. First, ML gets the past data of companies on metrics measuring companies’ operational performance, solvency, liquidity, asset efficiency, and debt repayments. Second, ML cleans the collected data by converting the data into a standard format that can be directly operated by computer programmes. Third, ML classifies data into distinct groups and tunes model parameters when ML learns from data. Fourth, the model accuracy is tested by SMEs’ data that have not been analysed by the model in the previous steps. Fifth, ML uses new data to improve the accuracy of an SME’s credit assessment by reducing errors.

There are three broad categories of ML algorithms and all of them have potential applications to SMEs’ credit assessment ([Buchanan and Miller, 2017](#)). First, supervised learning algorithms provide an output that is clearly defined in advance, e.g., a prediction of an SME’s credit score. We inform ML the definition of credit score and then supervise its learning procedure ([Lessmann, Baesens, Seow and Thomas, 2015](#)). Second, unsupervised learning algorithms automatically cluster data and find underlying structures or patterns of SMEs’ credit profiles. Third, reinforcement learning learns about data over time and

adjusts the way of assessing an SME's credit profile to maximize its estimation accuracy.

Python programming language is applied to implement ML and DL algorithms for SMEs' credit assessment. Python has the advantages of flexibility and efficiency as a high-level language as well as the advantage of fast computation through encapsulating functions written in a low-level language like C language. Among diverse function libraries, there are special libraries for machine learning. For example, the Scikit-Learn library provides convenient interfaces for basic ML tasks of supervised learning and unsupervised learning. The TensorFlow library that is originally developed by Google uses an intuitive flow structure for more advanced ML and DL design. These libraries are based on the fundamental Numpy and Scipy libraries for fast linear algebra operations and Matplotlib library for rich plotting styles.

A thorough and precise credit risk assessment is crucial for risk management and efficient allocation of financial resources to businesses. Various statistical methods have been proposed and employed by finance academics and professionals to measure the credit risk exposure of the businesses. In the recent years, due to the availability of big data, the more modern approaches such as machine learning (ML) and deep learning (DL) have risen in popularity. [Wallis, Kumar and Gepp \(2019\)](#) draw a comparison between the traditional statistical models and the modern ML techniques in terms of their performance in company credit scoring and find that the ML techniques outperform their traditional counterparts substantially. Also, their results show that among the various ML techniques, Random Forests tend to generate the most accurate results. The following paragraphs provide a brief description of some of the popular ML algorithms and their applications within the credit risk analysis literature.

One of the widely used algorithms in ML applications is the K-Nearest Neighbours (K-NN) algorithm developed by [Altman \(1992\)](#). This method makes predictions based on the position and the distance of a new observation relative to the training data set. For instance, if on a principal components plot, the majority of the K nearest observations to a new observation belong to a certain category, the new observation is classified in the same category. [Abdelmoula \(2015\)](#), [Marinakis, Marinaki, Doumpos, Matsatsinis and Zopounidis \(2008\)](#) and [Mukid, Widiharhi, Rusgiyono and Prahutama \(2018\)](#) are among the studies that employ K-NN models to estimate credit risk. Within the same research area, [Galindo and Tamayo \(2000\)](#) conduct a broad comparison of more than 9000 models and find that K-NN algorithm ranks among the top three performers.

Introduced by [Zou and Hastie \(2005\)](#) Elastic Net regression is another technique used in ML algorithms, which is based on a generalisation of the LASSO and Ridge regularised regressions. The LASSO and Ridge regressions both use similar methods to penalise for overfitting the data by introducing a penalty term for the regression parameters (namely, “L1” for LASSO and “L2” for Ridge regression). Hence, the higher the value of the regression parameters, the higher the penalty value. Based on this approach, the parameters are estimated by minimising the sum of squared residuals plus the penalty term. In the Elastic Net regression, the parameter estimations are done by incorporating both of the penalty terms in the model, which results in a more hybrid approach to model selection. [Gross and Siklos \(2020\)](#) employ the Elastic Net technique to reduce the dimensionality of their data while investigating the transmission of credit risk from the finance sector to the non-financial sector in Europe.

A large group of the most popular ML algorithms are based on the Decision

Tree algorithm. These include Random Forest, Extra Trees, Gradient Boost Machine and AdaBoost which are all ensemble models based on decision trees. In simple terms, the Decision Tree algorithm makes (regression or classification) predictions based on a decision algorithm that in multiple stages splits the data into two or more subgroups according to the available variables. The variables that have the highest power in explaining the variation in the data come first in this process. In a study comparing the performance of various ML and DL techniques and the Elastic Net regressions in credit risk assessment, [Addo, Guegan and Hassani \(2018\)](#) show that the tree-based models generally outperform their counterparts. Other comparative studies such as [Galindo and Tamayo \(2000\)](#), [Wallis et al. \(2019\)](#) and [Li, Mirza, Rahat and Xiong \(2020\)](#) also rank Decision Tree based models such as Random Forest among the top performers in application of credit risk analysis.

6 Empirical Results: PRI

We aim to estimate the following structural ordinary least squares (OLS) model:

$$\begin{aligned}
Post - Covid Solvency_{fsc} = & \alpha + \beta Pre - Covid Credit Score_{fsc} \\
& + \gamma_k \sum_{k=1}^K Firm - Level_{fk} + \delta_l \sum_{l=1}^L Regional_{rl} \\
& + \eta_m \sum_{m=1}^M Sectoral_{sm} + \lambda_n \sum_{n=1}^N Country - Level_{cn} + \epsilon_{fsc}
\end{aligned} \tag{1}$$

Where *Post - Covid Solvency_{fsc}* denotes a measure of post-covid solvency of firm *f*, which is headquartered in region *r*, operating in sector *s*, and located in country *c*. α is a constant term. *Pre - Covid Credit Score* is the firm's credit score before 1st April 2020. *Firm - Level* is a vec-

tor of firm-level variables and indices, *Regional* is a vector of regional variables and indices, *Sectoral* is a vector of sectoral variables and indices, and *Country – Level* is a vector of country-level variables and indices. ϵ is the error term. Heteroskedasticity-robust standard errors are used.

6.1 Determinants of Post-Covid Credit Scores

By estimating this model, we aim to predict the post-covid credit scores of UK firms using firm-level, regional, sectoral, and country-level indices. Table 9 reports the results of the OLS regressions where the dependent variable is *Post-Covid Credit Score*. Panel A reports the results for all UK firms and Panel B reports the results only for UK SMEs. Variable definitions are presented in Table 2.

We start by discussing the results for all UK firms (Table 9, Panel A). In Model (1), one standard deviation increase in *Pre-Covid Credit Score* would increase *Post-Covid Credit Score* by 20.66 points (significant at the 1% level). In Model (2), one standard deviation increase in *Firm-Level Financial Index* would increase *Post-Covid Credit Score* by 2.05 points (significant at the 1% level), and one standard deviation increase in *Firm-Level Non-Financial Index* would increase *Post-Covid Credit Score* by 3.59 points (significant at the 1% level). In Model (3), one standard deviation increase in *Firm-Level Overall Index* would increase *Post-Covid Credit Score* by 3.36 points (significant at the 1% level). In Model (4), one standard deviation increase in *Regional Covid Cases* would decrease *Post-Covid Credit Score* by 2.94 points (significant at the 1% level), one standard deviation increase in *Regional Covid Social Index* would increase *Post-Covid Credit Score* by 0.86 points (significant at the 1% level), and one standard deviation increase in *Regional GDHI Per*

Head would decrease *Post-Covid Credit Score* by 2.31 points (significant at the 1% level). In Models (5) and (6), *Sectoral Covid Exposure Index* and *Country-Level Stringency Index* are insignificant.

In Models (7) and (8), the results are qualitatively similar to Models (1) - (6) except that: a) the coefficient of *Regional Covid Social Index* becomes smaller and insignificant, b) the coefficient of *Sectoral Covid Exposure Index* becomes bigger and significant at the 1% level (In Model (7), one standard deviation increase in *Sectoral Covid Exposure Index* would decrease *Post-Covid Credit Score* by 1.40 points), and c) the coefficient of *Country-Level Stringency Index* becomes bigger and significant at the 1% level (In Model (8), one standard deviation increase in *Country-Level Stringency Index* would decrease *Post-Covid Credit Score* by 1.33 points).

The results for UK SMEs (Table 9, Panel B) are qualitatively similar to the results for all UK firms.

6.2 Determinants of Post-Covid County Court Judgements

By estimating this model, we aim to predict the post-covid value of county court judgements (CCJs) of UK firms using firm-level, regional, sectoral, and country-level indices. Table 10 reports the results of the OLS regressions where the dependent variable is *Post-Covid County Court Judgements*. Panel A reports the results for all UK firms and Panel B reports the results only for UK SMEs. Variable definitions are presented in Table 2.

We start by discussing the results for all UK firms (Table 10, Panel A). In Model (1), one standard deviation increase in *Pre-Covid Credit Score* would decrease *Post-Covid CCJs* by 6.5% (significant at the 1% level). In Model (2),

one standard deviation increase in *Firm-Level Financial Index* would increase *Post-Covid CCJs* by 31.4% (significant at the 1% level), and one standard deviation increase in *Firm-Level Non-Financial Index* would increase *Post-Covid CCJs* by 10.3% (significant at the 1% level). In Model (3), one standard deviation increase in *Firm-Level Overall Index* would increase *Post-Covid CCJs* by 50.1% (significant at the 1% level). In Model (4), *Regional Covid Cases* and *Regional GDHI Per Head* are insignificant, while one standard deviation increase in *Regional Covid Social Index* would increase *Post-Covid CCJs* by 26.2% (significant at the 5% level). In Model (5), one standard deviation increase in *Sectoral Covid Exposure Index* would increase *Post-Covid CCJs* by 9.5% (significant at the 1% level). In Model (6), one standard deviation increase in *Country-Level Stringency Index* would decrease *Post-Covid CCJs* by 16.1% (significant at the 5% level).

In Models (7) and (8), the results are qualitatively similar to Models (1) - (6) except that: a) the coefficient of *Firm-Level Financial Index* becomes smaller and insignificant, b) In Model (7), *Regional Covid Cases* becomes significant at the 5% level (one standard deviation increase in *Regional Covid Cases* would decrease *Post-Covid CCJs* by 27.2%), and c) the coefficient of *Country-Level Stringency Index* becomes smaller and insignificant.

The results for UK SMEs (Table 10, Panel B) are qualitatively similar to the results for all UK firms. Two exceptions are: a) *Firm-Level Non-Financial Index* enters negative rather than positive (one standard deviation increase in *Firm-Level Non-Financial Index* would decrease *Post-Covid CCJs* by 25.6% at the 1% significance level), and b) *Sectoral Covid Exposure Index* becomes insignificant in Models (7) and (8).

6.3 Determinants of Post-Covid Unsatisfied County Court Judgements

By estimating this model, we aim to predict the post-covid value of unsatisfied county court judgements (CCJs) of UK firms using firm-level, regional, sectoral, and country-level indices. Table 11 reports the results of the OLS regressions where the dependent variable is *Post-Covid Unsatisfied County Court Judgements*. Panel A reports the results for all UK firms and Panel B reports the results only for UK SMEs. Variable definitions are presented in Table 2.

Overall, the results reported for *Post-Covid Unsatisfied County Court Judgements* (Table 11) are qualitatively similar to those reported for *Post-Covid County Court Judgements* (Table 10). However, we still discuss the results in Table 11 below to show their statistical significance and economic magnitude.

We start by discussing the results for all UK firms (Table 11, Panel A). In Model (1), one standard deviation increase in *Pre-Covid Credit Score* would decrease *Post-Covid Unsatisfied CCJs* by 8.3% (significant at the 1% level). In Model (2), one standard deviation increase in *Firm-Level Financial Index* would increase *Post-Covid Unsatisfied CCJs* by 26.4% (significant at the 1% level), and one standard deviation increase in *Firm-Level Non-Financial Index* would increase *Post-Covid Unsatisfied CCJs* by 10.2% (significant at the 1% level). In Model (3), one standard deviation increase in *Firm-Level Overall Index* would increase *Post-Covid Unsatisfied CCJs* by 43.8% (significant at the 1% level). In Model (4), *Regional Covid Cases* and *Regional GDHI Per Head* are insignificant, while one standard deviation increase in *Regional Covid Social Index* would increase *Post-Covid Unsatisfied CCJs* by 22.6% (significant at the 5% level). In Model (5), one standard deviation increase in *Sectoral Covid Exposure Index* would increase *Post-Covid Unsatisfied CCJs* by 7.8%

(significant at the 1% level). In Model (6), one standard deviation increase in *Country-Level Stringency Index* would decrease *Post-Covid Unsatisfied CCJs* by 14.6% (significant at the 10% level).

In Models (7) and (8), the results are qualitatively similar to Models (1) - (6) except that: a) the coefficient of *Firm-Level Financial Index* becomes smaller and insignificant, b) In Model (7), *Regional Covid Cases* becomes significant at the 10% level (one standard deviation increase in *Regional Covid Cases* would decrease *Post-Covid Unsatisfied CCJs* by 27.2%), and c) the coefficient of *Country-Level Stringency Index* becomes smaller and insignificant.

The results for UK SMEs (Table 11, Panel B) are qualitatively similar to the results for all UK firms. Two exceptions are: a) *Firm-Level Non-Financial Index* enters negative rather than positive (one standard deviation increase in *Firm-Level Non-Financial Index* would decrease *Post-Covid Unsatisfied CCJs* by 21.5% at the 1% significance level), and b) *Sectoral Covid Exposure Index* becomes insignificant in Models (7) and (8).

Table 4. Firm-Level Financial Index
Panel A: Correlation Matrix

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
(1)	1.00																			
(2)	0.17	1.00																		
(3)	-0.01	0.71	1.00																	
(4)	0.00	-0.23	-0.16	1.00																
(5)	-0.01	-0.27	-0.19	0.25	1.00															
(6)	0.04	-0.15	-0.08	0.46	0.30	1.00														
(7)	0.10	0.03	0.00	0.06	0.00	0.06	1.00													
(8)	-0.10	0.07	0.05	-0.23	-0.71	-0.29	-0.05	1.00												
(9)	-0.06	0.03	0.05	-0.11	-0.34	-0.14	-0.02	0.48	1.00											
(10)	0.68	0.05	0.00	0.14	0.08	0.18	0.14	-0.14	-0.08	1.00										
(11)	0.10	-0.04	-0.06	0.11	0.03	0.06	0.08	-0.03	-0.05	0.15	1.00									
(12)	0.00	-0.01	0.01	0.03	0.03	0.01	0.00	-0.06	0.04	0.02	0.00	1.00								
(13)	-0.02	0.04	0.01	-0.13	-0.10	-0.11	-0.03	0.14	0.01	-0.09	-0.04	-0.83	1.00							
(14)	-0.11	-0.04	0.00	-0.03	-0.01	-0.03	-0.06	0.04	0.03	-0.14	-0.08	0.02	0.02	1.00						
(15)	-0.05	-0.11	-0.08	-0.02	0.11	-0.06	-0.02	-0.07	0.00	-0.04	-0.04	0.19	-0.15	0.03	1.00					
(16)	-0.03	-0.08	-0.09	-0.13	-0.29	-0.14	-0.03	0.33	0.16	-0.08	-0.05	-0.06	0.10	0.03	-0.02	1.00				
(17)	0.15	0.14	-0.03	-0.11	-0.08	-0.09	-0.02	0.05	-0.02	-0.05	-0.03	-0.45	0.48	0.01	-0.06	0.16	1.00			
(18)	0.04	-0.03	-0.05	-0.01	0.14	-0.03	-0.01	-0.24	-0.11	-0.02	-0.01	0.01	-0.03	0.01	0.00	0.00	0.05	1.00		
(19)	0.13	-0.07	-0.11	0.60	0.10	0.26	0.06	-0.14	-0.09	0.10	0.10	-0.05	0.02	-0.04	-0.07	-0.07	0.06	0.00	1.00	
(20)	0.15	0.00	-0.04	0.20	0.12	0.52	0.05	-0.15	-0.10	0.10	0.05	-0.20	0.18	-0.04	-0.08	-0.08	0.18	-0.01	0.43	1.00

This table shows the correlation coefficients of the variables used to construct the Firm-Level Financial Index. The variables are:

(1) Total Assets, (2) Fixed Assets%, (3) Tangible Assets%, (4) Debtors%, (5) Current Liabilities%, (6) Creditors%, (7) Balance Sheet Minorities%, (8) Shareholders Funds%, (9) Total Reserves%, (10) Turnover, (11) Overseas Turnover%, (12) EBIT%, (13) Administration Expenses%, (14) Exceptional Items%, (15) ROA, (16) Liquidity Ratio, (17) Assets Turnover, (18) Gearing, (19) Debtors Collection Days, and (20) Creditors Payment Days. Variable definitions are presented in Table 2.

Table 4: Firm-Level Financial Index
Panel B: Kaiser-Meyer-Olkin (KMO) Test for Sampling Adequacy

VARIABLE	KMO
Total Assets	0.4987
Fixed Assets%	0.5366
Tangible Assets%	0.5261
Debtors%	0.5704
Current Liabilities%	0.7024
Creditors%	0.589
Balance Sheet Minorities%	0.8086
Shareholders Funds%	0.6655
Total Reserves%	0.794
Turnover	0.5364
Overseas Turnover%	0.7815
EBIT%	0.596
Administration Expenses%	0.6076
Exceptional Items%	0.7849
ROA	0.7604
Liquidity Ratio	0.8018
Assets Turnover	0.7512
Gearing	0.6093
Debtors Collection Days	0.523
Creditors Payment Days	0.5296
Overall	0.6012

This table shows the Kaiser-Meyer-Olkin (KMO) test for sampling adequacy for the Firm-Level Financial Index. Variable definitions are presented in Table 2.

Table 4: Firm-Level Financial Index
Panel C: Eigenvalues of the Components

Component	Eigenvalue	Difference	Proportion	Cumulative
COMP1	3.1435	0.7327	0.1572	0.1572
COMP2	2.4108	0.4646	0.1205	0.2777
COMP3	1.9462	0.2515	0.0973	0.3750
COMP4	1.6947	0.2301	0.0847	0.4598
COMP5	1.4646	0.4031	0.0732	0.5330
COMP6	1.0615	0.0679	0.0531	0.5861
COMP7	0.9935	0.0460	0.0497	0.6357
COMP8	0.9475	0.0155	0.0474	0.6831
COMP9	0.9320	0.0267	0.0466	0.7297
COMP10	0.9053	0.0741	0.0453	0.7750
COMP11	0.8312	0.0361	0.0416	0.8165
COMP12	0.7951	0.1302	0.0398	0.8563
COMP13	0.6649	0.0581	0.0332	0.8895
COMP14	0.6068	0.1081	0.0303	0.9199
COMP15	0.4987	0.2217	0.0249	0.9448
COMP16	0.2770	0.0361	0.0138	0.9587
COMP17	0.2409	0.0135	0.0120	0.9707
COMP18	0.2274	0.0205	0.0114	0.9821
COMP19	0.2069	0.0554	0.0103	0.9924
COMP20	0.1515	.	0.0076	1.0000

This table shows the eigenvalues of the components of the Firm-Level Financial Index.

Table 4: Firm-Level Financial Index
Panel D: Loadings of the Variables

VARIABLE	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6
Total Assets	0.1201	0.1509	0.4579	0.1012	-0.4161	0.2470
Fixed Assets%	-0.1837	0.0866	0.4949	-0.2257	0.3177	0.0804
Tangible Assets%	-0.1658	0.0129	0.4239	-0.2350	0.4348	0.0051
Debtors%	0.3615	0.0422	-0.0762	0.2839	0.2273	0.0111
Current Liabilities%	0.3857	-0.0631	-0.1482	-0.3342	-0.0774	-0.0586
Creditors%	0.3647	0.0922	-0.0026	0.1550	0.2539	0.0584
Balance Sheet Minorities%	0.0739	0.0316	0.1581	0.0933	-0.0840	-0.3405
Shareholders Funds%	-0.3967	0.0417	-0.0137	0.4202	0.0445	-0.0199
Total Reserves%	-0.2541	-0.0376	0.0009	0.3674	0.0726	0.0689
Turnover	0.1947	0.0766	0.4387	0.1521	-0.3831	0.0689
Overseas Turnover%	0.1065	0.0304	0.1027	0.1556	-0.1340	-0.4881
EBIT%	0.0734	-0.5354	0.1548	0.1369	0.0484	0.1864
Administration Expenses%	-0.1422	0.5262	-0.1646	-0.1443	-0.0492	-0.1364
Exceptional Items%	-0.0597	-0.0422	-0.1662	-0.0510	0.1313	0.3827
ROA	0.0270	-0.1986	-0.0960	-0.0574	-0.1327	0.2120
Liquidity Ratio	-0.2124	0.0730	-0.1298	0.2609	-0.1866	0.2896
Assets Turnover	-0.0970	0.4347	-0.0499	-0.1169	-0.1272	0.2060
Gearing	0.0821	-0.0121	-0.0406	-0.2610	-0.1684	0.3594
Debtors Collection Days	0.2889	0.1990	0.0007	0.2956	0.2419	0.1585
Creditors Payment Days	0.2463	0.3248	0.0294	0.1306	0.2415	0.1903

This table shows the loadings of the variables in the Firm-Level Financial Index. The loadings are shown only for the components whose eigenvalues exceed 1. The loadings exceeding 0.30 are gray-highlighted. Variable definitions are presented in Table 2.

Table 5. Firm-Level Non-Financial Index
Panel A: Correlation Matrix

VARIABLES	(21)	(22)	(23)	(24)	(25)	(26)	(27)
(21)	1.00						
(22)	-0.16	1.00					
(23)	-0.19	0.07	1.00				
(24)	-0.12	0.16	0.05	1.00			
(25)	0.09	0.12	0.02	0.01	1.00		
(26)	-0.04	0.09	0.07	0.03	0.05	1.00	
(27)	-0.14	0.21	0.10	0.39	0.01	0.06	1.00

This table shows the correlation coefficients of the variables in the Firm-Level Non-Financial Index. The variables are: (21) SME, (22) Listed Firm, (23) Firm Age, (24) Number of Employees, (25) Firm Independence, (26) Business Diversification, and (27) Geographical Diversification. Variable definitions are presented in Table 2.

Table 5: Firm-Level Non-Financial Index
Panel B: Kaiser-Meyer-Olkin (KMO) Test for Sampling Adequacy

VARIABLES	KMO
SME	0.6016
Listed Firm	0.6505
Firm Age	0.5941
Number of Employees	0.5819
Firm Independence	0.4396
Business Diversification	0.6668
Geographical Diversification	0.5868
Overall	0.593

This table shows the Kaiser-Meyer-Olkin (KMO) test for sampling adequacy for the Firm-Level Non-Financial Index. Variable definitions are presented in Table 2.

Table 5: Firm-Level Non-Financial Index
Panel C: Eigenvalues of the Components

Components	Eigenvalue	Difference	Proportion	Cumulative
Comp1	1.6948	0.5841	0.2421	0.2421
Comp2	1.1107	0.0307	0.1587	0.4008
Comp3	1.0801	0.1506	0.1543	0.5551
Comp4	0.9295	0.0655	0.1328	0.6879
Comp5	0.8640	0.1424	0.1234	0.8113
Comp6	0.7216	0.1223	0.1031	0.9144
Comp7	0.5993	.	0.0856	1.0000

This table shows the eigenvalues of the components of the Firm-Level Non-Financial Index.

Table 5: Firm-Level Non-Financial Index
Panel D: Loadings of the Variables

VARIABLES	Comp1	Comp2	Comp3
SME	-0.3787	0.4729	-0.2789
Listed Firm	0.4285	0.2892	0.1136
Firm Age	0.2825	-0.295	0.5469
Number of Employees	0.5057	0.04375	-0.4494
Firm Independence	0.06176	0.7358	0.2404
Business Diversification	0.1846	0.2456	0.4848
Geographical Diversification	0.5473	0.04677	-0.3396

This table shows the loadings of the variables in the Firm-Level Non-Financial Index. The loadings are shown only for the components whose eigenvalues exceed 1. The loadings exceeding 0.30 are gray-highlighted. Variable definitions are presented in Table 2.

Table 6. Firm-Level Overall Index

Panel A: Correlation Matrix

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
(21)	-0.49	-0.14	-0.04	-0.10	-0.05	-0.13	-0.12	0.12	0.07	-0.60	-0.12	0.00	0.08	0.12	0.07	0.07	-0.05	0.02	-0.09	-0.10
(22)	0.27	0.11	-0.02	-0.01	-0.05	-0.01	0.10	0.02	0.00	0.20	0.12	-0.05	0.05	-0.19	-0.03	0.00	0.07	-0.01	0.06	0.08
(23)	0.14	0.10	0.06	0.08	-0.08	0.09	0.01	0.10	0.05	0.18	0.09	0.00	-0.04	-0.04	-0.12	-0.03	0.01	-0.04	0.10	0.07
(24)	0.24	0.04	0.02	0.01	0.01	0.03	0.09	-0.04	-0.02	0.25	0.03	0.00	-0.01	-0.07	-0.01	-0.02	0.00	0.00	0.03	0.04
(25)	-0.09	0.18	0.21	-0.18	-0.32	-0.15	-0.01	0.33	0.22	-0.13	-0.14	-0.04	0.19	0.03	-0.10	0.13	-0.03	-0.05	-0.15	-0.09
(26)	0.04	0.01	0.00	0.01	-0.03	0.03	0.04	0.03	0.01	0.07	0.02	-0.02	0.01	-0.03	-0.03	0.00	0.00	-0.01	0.01	0.02
(27)	0.28	0.04	0.00	0.00	0.00	0.03	0.08	-0.03	-0.01	0.26	0.03	0.00	0.00	-0.08	-0.01	-0.01	0.02	-0.01	0.04	0.07

This table shows the correlation coefficients of the variables used to construct the Firm-Level Overall Index. The variables are:

- (1) Total Assets, (2) Fixed Assets%, (3) Tangible Assets%, (4) Debtors%, (5) Current Liabilities%, (6) Creditors%, (7) Balance Sheet Minorities%, (8) Shareholders Funds%, (9) Total Reserves%, (10) Turnover, (11) Overseas Turnover%, (12) EBIT%, (13) Administration Expenses%, (14) Exceptional Items%, (15) ROA, (16) Liquidity Ratio, (17) Assets Turnover, (18) Gearing, (19) Debtors Collection Days, (20) Creditors Payment Days, (21) SME, (22) Listed Firm, (23) Firm Age, (24) Number of Employees, (25) Firm Independence, (26) Business Diversification, and (27) Geographical Diversification. Variable definitions are presented in

Table 2.

Table 8. Sectoral Covid Exposure Index

Sectoral Covid Exposure Index	Sector Name	Two-Digit SIC Code	MARs(-5,-4)
66	Local and interurban passenger transit	41	-46.09%
65	Eating and drinking places	58	-28.21%
64	Apparel and accessory stores	56	-24.85%
63	Amusement & recreation services	79	-24.47%
62	Miscellaneous Services	89	-23.56%
61	Transportation by air	45	-22.80%
60	Hotels and other lodging places	70	-21.05%
59	Apparel and other textile products	23	-20.81%
58	Water transportation	44	-20.66%
57	Textile mill products	22	-19.60%
56	Motion pictures	78	-19.34%
55	Auto repair, services, and parking	75	-19.10%
54	Transportation services	47	-16.78%
53	Wholesale trade-nondurable goods	51	-16.74%
52	Building materials & garden supplies	52	-16.55%
51	Lumber and wood products	24	-16.51%
50	Heavy construction, except building	16	-16.32%
49	Miscellaneous retail	59	-15.23%

Table 8: Sectoral Covid Exposure Index (continued)

Sectoral Covid Exposure Index	Sector Name	Two-Digit SIC Code	MARs(-5,-4)
48	Miscellaneous manufacturing	39	-15.20%
47	Nondepository institutions	61	-14.79%
46	Paper and allied products	26	-14.12%
45	Depository institutions	60	-13.22%
44	Real estate	65	-13.02%
43	Oil and gas extraction	13	-12.81%
42	Wholesale trade-durable goods	50	-12.80%
41	Automotive dealers & service stations	55	-12.78%
40	Transportation equipment	37	-12.76%
39	Electronic & other electric equipment	36	-12.50%
38	Leather and leather products	31	-12.27%
37	Special trade contractors	17	-11.46%
36	Furniture and homefurnishings stores	57	-11.43%
35	General contractors & operative builders	15	-11.31%
34	Industrial machinery and equipment	35	-11.31%
33	Printing and publishing	27	-11.20%

Table 8: Sectoral Covid Exposure Index (continued)

Sectoral Covid Exposure Index	Sector Name	Two-Digit SIC Code	MARs(-5,-4)
32	Food and kindred products	20	-11.11%
31	Legal services	81	-10.78%
30	Agricultural Production - Crops	1	-10.08%
29	Stone, clay, and glass products	32	-10.03%
28	Fabricated metal products	34	-9.97%
27	Trucking and warehousing	42	-9.74%
26	Chemicals and allied products	28	-9.61%
25	Instruments and related products	38	-9.16%
24	Business services	73	-9.12%
23	Security and commodity brokers	62	-8.01%
22	Coal mining	12	-7.58%
21	Rubber and misc. plastics products	30	-7.28%
20	Metal mining	10	-7.26%
19	Health services	80	-6.80%
18	Holding and other investment offices	67	-6.54%
17	Food stores	54	-6.13%
16	Agricultural services	7	-5.30%
15	Communication	48	-4.92%
14	Petroleum and coal products	29	-4.61%
13	Engineering & management services	87	-4.15%

Table 8: Sectoral Covid Exposure Index (continued)

Sectoral Covid Exposure Index	Sector Name	Two-Digit SIC Code	MARs(-5,-4)
12	Furniture and fixtures	25	-3.57%
11	Forestry	8	-3.51%
10	Primary metal industries	33	-3.38%
9	Personal services	72	-3.12%
8	Insurance carriers	63	-2.83%
7	Electric, gas, and sanitary services	49	-2.02%
6	Educational services	82	-1.98%
5	Nonmetallic minerals, except fuels	14	-0.58%
4	Tobacco products	21	1.40%
3	General merchandise stores	53	1.86%
2	Insurance agents, brokers, & service	64	1.89%
1	Nonclassified Sectors	99	3.15%

This table shows the scores of the Sectoral Covid Exposure Index.

Table 6: Firm-Level Overall Index
Panel B: Kaiser-Meyer-Olkin (KMO) Test for Sampling Adequacy

VARIABLES	KMO
Total Assets	0.6854
Fixed Assets%	0.5766
Tangible Assets%	0.5685
Debtors%	0.5925
Current Liabilities%	0.7166
Creditors%	0.6102
Balance Sheet Minorities%	0.829
Shareholders Funds%	0.6827
Total Reserves%	0.8571
Turnover	0.681
Overseas Turnover%	0.7782
EBIT%	0.602
Administration Expenses%	0.6101
Exceptional Items%	0.7907
ROA	0.739
Liquidity Ratio	0.8259
Assets Turnover	0.7187
Gearing	0.647
Debtors Collection Days	0.5418
Creditors Payment Days	0.546
SME	0.8273
Listed Firm	0.738
Firm Age	0.7877
Number of Employees	0.7267
Firm Independence	0.7802
Business Diversification	0.7385
Geographical Diversification	0.7427
Overall	0.6575

This table shows the Kaiser-Meyer-Olkin (KMO) test for sampling adequacy for the Firm-Level Overall Index. Variable definitions are presented in Table 2.

Table 6: Firm-Level Overall Index
Panel C: Eigenvalues of the Components

Components	Eigenvalue	Difference	Proportion	Cumulative
Comp1	3.6738	0.8263	0.1361	0.1361
Comp2	2.8475	0.4702	0.1055	0.2415
Comp3	2.3774	0.6556	0.0881	0.3296
Comp4	1.7218	0.1109	0.0638	0.3934
Comp5	1.6108	0.3826	0.0597	0.4530
Comp6	1.2283	0.1229	0.0455	0.4985
Comp7	1.1054	0.0882	0.0409	0.5394
Comp8	1.0172	0.0243	0.0377	0.5771
Comp9	0.9929	0.0067	0.0368	0.6139
Comp10	0.9862	0.0217	0.0365	0.6504
Comp11	0.9645	0.0234	0.0357	0.6861
Comp12	0.9410	0.0999	0.0349	0.7210
Comp13	0.8411	0.0284	0.0312	0.7521
Comp14	0.8127	0.0154	0.0301	0.7822
Comp15	0.7973	0.0320	0.0295	0.8118
Comp16	0.7654	0.0748	0.0283	0.8401
Comp17	0.6906	0.0761	0.0256	0.8657
Comp18	0.6145	0.0246	0.0228	0.8885
Comp19	0.5900	0.0264	0.0219	0.9103
Comp20	0.5636	0.0576	0.0209	0.9312
Comp21	0.5059	0.0684	0.0187	0.9499
Comp22	0.4375	0.2115	0.0162	0.9661
Comp23	0.2260	0.0364	0.0084	0.9745
Comp24	0.1897	0.0151	0.0070	0.9815
Comp25	0.1746	0.0080	0.0065	0.9880
Comp26	0.1665	0.0086	0.0062	0.9942
Comp27	0.1579	.	0.0058	1.0000

This table shows the eigenvalues of the components of the Firm-Level Overall Index.

Table 6: Firm-Level Overall Index
Panel D: Loadings of the Variables

VARIABLES	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6	Comp7	Comp8
Total Assets	0.1981	0.3886	-0.1412	-0.0179	-0.1600	-0.1965	0.1771	0.0312
Fixed Assets%	-0.1564	0.3035	-0.0978	-0.3656	0.3756	0.0036	0.0161	0.1293
Tangible Assets%	-0.1696	0.2007	-0.1172	-0.3590	0.4546	0.0752	0.0482	0.0867
Debtors%	0.3153	-0.1148	0.0749	0.2756	0.2479	0.0515	0.0090	0.0546
Current Liabilities%	0.3544	-0.2383	0.0427	-0.2505	-0.1067	0.0866	-0.0216	-0.0972
Creditors%	0.3255	-0.0553	0.0950	0.1210	0.2906	0.1156	0.1162	-0.0754
Balance Sheet Minorities%	0.0828	0.1074	-0.0577	0.0404	-0.0101	0.1230	-0.2203	0.2727
Shareholders Funds%	-0.3552	0.1495	-0.0029	0.3974	0.0447	-0.0722	0.0229	0.0385
Total Reserves%	-0.2288	0.0696	-0.0842	0.3492	0.0808	-0.0231	0.1348	0.0950
Turnover	0.2606	0.3286	-0.2118	0.0137	-0.1259	-0.2147	0.1197	-0.0945
Overseas Turnover%	0.1141	0.0714	-0.0328	0.1277	-0.0391	-0.1957	-0.4975	0.1256
EBIT%	0.0199	-0.1911	-0.5106	0.1029	0.1010	0.0141	0.0796	0.0825
Administration Expenses%	-0.1111	0.2007	0.5093	-0.1056	-0.1332	0.0300	-0.0279	-0.0373
Exceptional Items%	-0.0779	-0.1285	0.0558	0.0016	0.0558	-0.0700	0.4650	-0.2944
ROA	0.0028	-0.1627	-0.1327	-0.0460	-0.2292	0.0335	0.1933	0.3434
Liquidity Ratio	-0.1704	0.0469	0.0774	0.2858	-0.2031	-0.1048	0.1530	0.0166
Assets Turnover	-0.0253	0.2035	0.3808	-0.0961	-0.1603	-0.1552	0.1238	0.1112
Gearing	0.0490	-0.0541	0.0209	-0.2061	-0.1299	0.0163	-0.0131	-0.1167
Debtors Collection Days	0.2741	0.0345	0.1891	0.2680	0.2679	0.0288	0.0522	0.2241
Creditors Payment Days	0.2399	0.1218	0.3108	0.0821	0.2375	0.0927	0.1596	0.1276
SME	-0.2058	-0.3006	0.1602	0.0243	0.0295	0.3463	-0.0578	0.0668
Listed Firm	0.0628	0.2594	-0.0473	0.0721	-0.1356	0.2620	-0.3332	0.1677
Firm Age	0.0565	0.1740	-0.0613	0.1601	0.2037	-0.1607	-0.0565	-0.3926
Number of Employees	0.0919	0.2085	-0.1190	0.0301	-0.1870	0.4635	0.2242	-0.0012
Firm Independence	-0.2410	0.1217	0.0184	0.0868	0.1226	0.3240	-0.0409	-0.0981
Business Diversification	0.0155	0.0866	-0.0182	0.0848	-0.0009	0.1924	-0.2951	-0.5849
Geographical Diversification	0.1019	0.2359	-0.1139	0.0544	-0.1925	0.4515	0.1963	-0.0567

This table shows the loadings of the variables in the Firm-Level Overall Index. The loadings are shown only for the components whose eigenvalues exceed 1. The loadings exceeding 0.30 are gray-highlighted. Variable definitions are presented in Table 2.

Table 7. Regional Covid Social Index
Panel A: Correlation Matrix

VARIABLES	Severe Anxiety	Covid Information	Self-isolation	Affordability	Pessimism
Severe Anxiety	1.00				
Covid Information	-0.86	1.00			
Self-isolation	0.72	-0.73	1.00		
Affordability	-0.48	0.56	-0.61	1.00	
Pessimism	0.41	-0.63	0.67	-0.55	1.00

This table shows the correlation coefficients of the variables used to construct the Regional Covid Social Index. Variable definitions are presented in Table 2.

Table 7: Regional Covid Social Index
Panel B: Kaiser-Meyer-Olkin (KMO) Test for Sampling Adequacy

VARIABLES	KMO
Severe Anxiety	0.6163
Covid Information	0.6927
Self-isolation	0.7934
Affordability	0.9229
Pessimism	0.6451
Overall	0.714

This table shows the Kaiser-Meyer-Olkin (KMO) test for sampling adequacy for the Regional Covid Social Index. Variable definitions are presented in Table 2.

Table 7: Regional Covid Social Index
Panel C: Eigenvalues of the Components

Components	Eigenvalue	Difference	Proportion	Cumulative
Comp1	3.5122	2.8413	0.7024	0.7024
Comp2	0.6709	0.2039	0.1342	0.8366
Comp3	0.4670	0.2041	0.0934	0.9300
Comp4	0.2629	0.1759	0.0526	0.9826
Comp5	0.0870	.	0.0174	1.0000

This table shows the eigenvalues of the components of the Regional Covid Social Index.

Table 7: Regional Covid Social Index
Panel D: Loadings of the Variables

VARIABLES	Comp1
Severe Anxiety	0.4483
Covid Information	-0.4871
Self-isolation	0.4787
Affordability	-0.4023
Pessimism	0.4131

This table shows the loadings of the variables in the Regional Covid Social Index. The loadings are shown only for the components whose eigenvalues exceed 1. The loadings exceeding 0.30 are gray-highlighted. Variable definitions are presented in Table 2.

Table 9. Determinants of Post-Covid Credit Scores
Panel A: All UK Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	0.7411*** (475.0345)						0.8375*** (127.9562)	0.8417*** (136.1038)
Firm-Level Financial Index		1.1573*** (20.3748)					0.3720*** (3.6345)	
Firm-Level Non-Financial Index		2.7552*** (9.6593)					0.3123*** (2.3304)	
Firm-Level Overall Index			1.7502*** (39.3996)					0.3576*** (3.8193)
Regional Covid Cases				-0.0037*** (-24.8717)			-0.0014*** (-5.4509)	-0.0014*** (-5.4549)
Regional Covid Social Index				0.4609*** (5.7365)			0.0857 (0.9335)	0.0902 (0.9844)
Regional GDHI Per Head				-0.0005*** (-13.3085)			-0.0002*** (-5.7744)	-0.0002*** (-5.7077)
Sectoral Covid Exposure Index					0.0028 (0.7003)		-0.0798*** (-7.4010)	-0.0774*** (-7.1575)
Country-Level Stringency Index						-0.0201 (-0.4865)	-0.3705*** (-6.2148)	-0.3778*** (-6.3291)
Constant	13.2188*** (149.2154)	68.6106*** (734.5634)	69.1608*** (760.3831)	59.4498*** (70.1959)	55.1644*** (396.7061)	47.8041*** (15.3723)	44.5529*** (9.4322)	44.7962*** (9.4280)
Observations	174,382	85,685	85,685	32,104	162,083	32,773	10,029	10,029
R-squared	0.5524	0.0340	0.0157	0.0163	0.0000	0.0000	0.6686	0.6681

This table shows the regression results regarding the determinants of post-covid credit scores in all UK firms. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

Table 9: Determinants of Post-Covid Credit Scores
Panel B: UK SMEs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	0.7408*** (435.7773)						0.8313*** (118.3523)	0.8455*** (132.5251)
Firm-Level Financial Index		1.3836*** (22.8665)					0.1254 (1.0417)	
Firm-Level Non-Financial Index		14.8813*** (37.3769)					2.2196*** (5.8730)	
Firm-Level Overall Index			1.1568*** (19.5677)					0.0319 (0.2680)
Regional Covid Cases				-0.0031*** (-21.1547)			-0.0013*** (-4.8014)	-0.0012*** (-4.7089)
Regional Covid Social Index				0.2669*** (3.3916)			0.0523 (0.5292)	0.0526 (0.5313)
Regional GDHI Per Head				-0.0004*** (-12.6649)			-0.0003*** (-6.4105)	-0.0003*** (-6.1551)
Sectoral Covid Exposure Index					0.0004 (0.0994)		-0.0602*** (-5.4284)	-0.0512*** (-4.6184)
Country-Level Stringency Index						0.0114 (0.2829)	-0.3106*** (-4.9056)	-0.2942*** (-4.6175)
Constant	13.0363*** (144.0223)	69.9066*** (512.2154)	67.1146*** (637.4233)	56.1387*** (67.0635)	52.4517*** (367.0388)	43.2034*** (14.2559)	41.0956*** (8.1467)	37.9036*** (7.4923)
Observations	153,663	69,476	69,476	29,604	141,808	30,252	8,346	8,346
R-squared	0.5562	0.0743	0.0051	0.0141	0.0000	0.0000	0.6817	0.6798

This table shows the regression results regarding the determinants of post-covid credit scores in UK SMEs. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

Table 10. Determinants of Post-Covid County Court Judgements
Panel A: All UK Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	-0.0024*** (-2.6835)						-0.0163*** (-2.5995)	-0.0146** (-2.3440)
Firm-Level Financial Index		0.1541*** (6.9929)					0.0336 (0.4495)	
Firm-Level Non-Financial Index		0.0755*** (4.1089)					0.1444*** (3.2699)	
Firm-Level Overall Index			0.2117*** (11.3249)					0.1744*** (2.7991)
Regional Covid Cases				0.0001 (0.4061)			-0.0004** (-2.1903)	-0.0002 (-1.3909)
Regional Covid Social Index				0.1241** (2.5474)			0.1827** (2.4999)	0.1812** (2.4518)
Regional GDHI Per Head				-0.0000 (-1.0179)			-0.0000 (-0.9474)	-0.0000 (-0.8792)
Sectoral Covid Exposure Index					0.0052*** (3.0440)		0.0182** (2.3420)	0.0181** (2.3085)
Country-Level Stringency Index						-0.0499** (-2.0726)	0.0122 (0.2975)	0.0149 (0.3619)
Constant	1.2435*** (19.8093)	0.9310*** (23.9424)	0.8986*** (24.2142)	1.4764*** (2.9447)	0.9895*** (15.6156)	4.9924*** (2.7432)	1.5394 (0.4606)	1.0895 (0.3268)
Observations	10,703	6,625	6,625	1,030	10,427	1,276	467	467
R-squared	0.0007	0.0253	0.0289	0.0075	0.0010	0.0035	0.0811	0.0657

This table shows the regression results regarding the determinants of post-covid county court judgements in all UK firms. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

Table 10: Determinants of Post-Covid County Court Judgements
Panel B: UK SMEs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	-0.0080*** (-8.3478)						-0.0197*** (-3.0416)	-0.0193*** (-3.0563)
Firm-Level Financial Index		0.0883*** (3.1664)					0.2140** (2.1183)	
Firm-Level Non-Financial Index		-0.2268*** (-3.1950)					0.0662 (0.5925)	
Firm-Level Overall Index			0.0973*** (3.5260)					0.2213** (2.1993)
Regional Covid Cases				0.0002 (1.3242)			-0.0002 (-0.6792)	-0.0002 (-0.6982)
Regional Covid Social Index				0.0992* (1.8654)			0.1683* (1.9639)	0.1683* (1.9618)
Regional GDHI Per Head				-0.0000 (-0.4625)			-0.0000 (-1.1053)	-0.0000 (-1.1180)
Sectoral Covid Exposure Index					0.0039** (2.1555)		-0.0046 (-0.5274)	-0.0052 (-0.5897)
Country-Level Stringency Index						-0.0612** (-2.5213)	-0.0312 (-0.6300)	-0.0335 (-0.6735)
Constant	1.2774*** (19.8255)	0.7947*** (18.4224)	0.8596*** (20.6203)	0.9895* (1.7059)	0.8053*** (12.0400)	5.5739*** (3.0340)	5.6738 (1.3932)	5.9061 (1.4389)
Observations	7,055	3,550	3,550	753	6,843	952	250	250
R-squared	0.0100	0.0054	0.0043	0.0092	0.0007	0.0066	0.0835	0.0844

This table shows the regression results regarding the determinants of post-covid county court judgements in UK SMEs. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

Table 11. Determinants of Post-Covid Unsatisfied County Court Judgements
Panel A: All UK Firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	-0.0031*** (-3.6478)						-0.0185*** (-3.0599)	-0.0169*** (-2.7990)
Firm-Level Financial Index		0.1322*** (6.1605)					0.0467 (0.6470)	
Firm-Level Non-Financial Index		0.0749*** (4.1883)					0.1479*** (3.4605)	
Firm-Level Overall Index			0.1895*** (10.3879)					0.1886*** (3.1327)
Regional Covid Cases				0.0001 (0.8229)			-0.0004* (-1.7882)	-0.0001 (-0.8864)
Regional Covid Social Index				0.1089** (2.3638)			0.1399** (2.0880)	0.1381** (2.0362)
Regional GDHI Per Head				-0.0000 (-0.8028)			-0.0000 (-0.9345)	-0.0000 (-0.8605)
Sectoral Covid Exposure Index					0.0043*** (2.6568)		0.0117* (1.6737)	0.0116 (1.6326)
Country-Level Stringency Index						-0.0446* (-1.9260)	0.0234 (0.6113)	0.0259 (0.6735)
Constant	1.1591*** (19.2286)	0.8009*** (21.6607)	0.7732*** (21.9545)	1.1973** (2.5256)	0.8871*** (14.6977)	4.4606** (2.5459)	0.7863 (0.2520)	0.3543 (0.1138)
Observations	10,703	6,625	6,625	1,030	10,427	1,276	467	467
R-squared	0.0014	0.0244	0.0260	0.0072	0.0008	0.0031	0.0844	0.0669

This table shows the regression results regarding the determinants of post-covid unsatisfied county court judgements in all UK firms. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

Table 11: Determinants of Post-Covid Unsatisfied County Court Judgements
Panel B: UK SMEs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-Covid Credit Score	-0.0082*** (-8.9391)						-0.0205*** (-3.3368)	-0.0199*** (-3.3183)
Firm-Level Financial Index		0.0860*** (3.1451)					0.1658 (1.6357)	
Firm-Level Non-Financial Index		-0.1855*** (-2.7049)					0.1249 (1.1195)	
Firm-Level Overall Index			0.0955*** (3.5292)					0.1818* (1.7875)
Regional Covid Cases				0.0003 (1.5205)			-0.0001 (-0.3532)	-0.0001 (-0.3574)
Regional Covid Social Index				0.1038** (2.0322)			0.1457* (1.8668)	0.1474* (1.8840)
Regional GDHI Per Head				-0.0000 (-0.6251)			-0.0001 (-1.5680)	-0.0001 (-1.5786)
Sectoral Covid Exposure Index					0.0032* (1.8760)		-0.0054 (-0.6754)	-0.0061 (-0.7589)
Country-Level Stringency Index						-0.0508** (-2.1578)	0.0013 (0.0275)	-0.0003 (-0.0070)
Constant	1.2096*** (19.4001)	0.7077*** (17.2253)	0.7640*** (19.3495)	0.9488* (1.6897)	0.7482*** (11.6196)	4.6948*** (2.6346)	3.5075 (0.9023)	3.6440 (0.9319)
Observations	7,055	3,550	3,550	753	6,843	952	250	250
R-squared	0.0114	0.0051	0.0045	0.0123	0.0005	0.0050	0.1007	0.1023

This table shows the regression results regarding the determinants of post-covid unsatisfied county court judgements in UK SMEs. Variable definitions are presented in Table 2. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, 10% level respectively.

7 Empirical results: Machine/deep learning methods in credit assessment

7.1 Sufficiency of sample size

Using the data of UK companies from FAME and other sources, we apply machine learning models to estimate firms' credit profiles. The research problem is to estimate firms' credit scores by applying regression models in machine learning. During estimation, we predict firms' credit scores and examine the plots about residuals and prediction errors. We also show that the sample size is sufficient for the model to achieve stable prediction scores. Fig. 2 shows the estimation accuracy of some machine learning methods. The results are reasonably well and the performance can be improved after the fine tune of parameters.

Recent references find that some popular boosting machine learning methods achieve superior performance in credit assessment. We also use three typical boosting methods (XGBoost, LightGBM, and CatBoost) to our data and find that they do achieve better performance, e.g., higher R^2 values, in predicting firms' credit scores. The R^2 values of three boosting methods are similar and LightGBM is much faster.

We carry out a comprehensive analysis of UK firms' credit scores in the 12 regions individually. We run some exploratory data analyses and find that most regions show two fat tails in credit scores except London that has a relatively even distribution. After that, we start from a variety of machine learning methods including popular boosting methods that recommended by recent references for credit assessment. These methods provide consistent high accuracies and stable performance. Then we fine-tune the model parameters

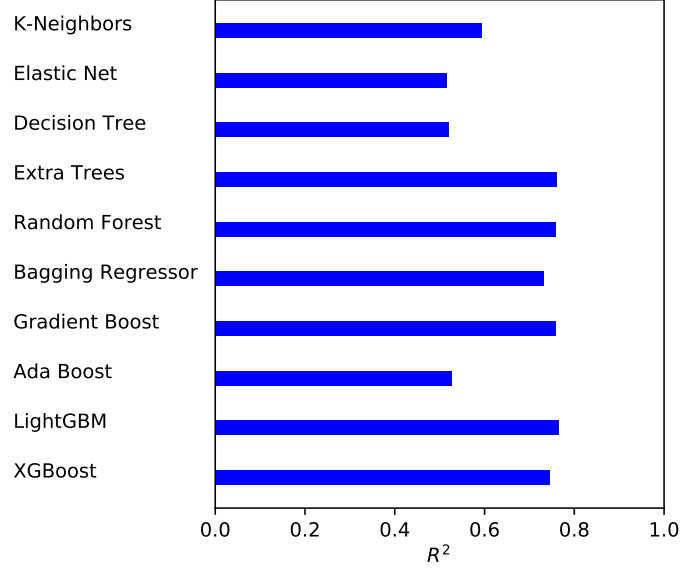


Figure 2. Estimation performance of different machine learning models.

of LightGBM boosting method and apply it to 12 regions, which all show robust and high performance on the plots about residuals and prediction errors. Besides, the plots about the importance of firm features for the 12 regions are similar, where the likelihood of failure (the firm age) has the most significant negative (positive) effects on credit scores. Finally, we obtain the residuals after estimating the credit scores of the 12 regions.

We examine the residuals of the estimates to evaluate whether the estimated machine learning models are adequate to describe the data. We calculate the residuals for the 12 regions in the UK individually and find that all of the residuals demonstrate series like white noises. Furthermore, using some classic information criteria to these residuals shows that their autoregressive orders are zeros. Both results indicate that the models estimated for the 12 regions are adequate descriptions of the regional data and there are not further

additional structures in the residuals of the estimated model.

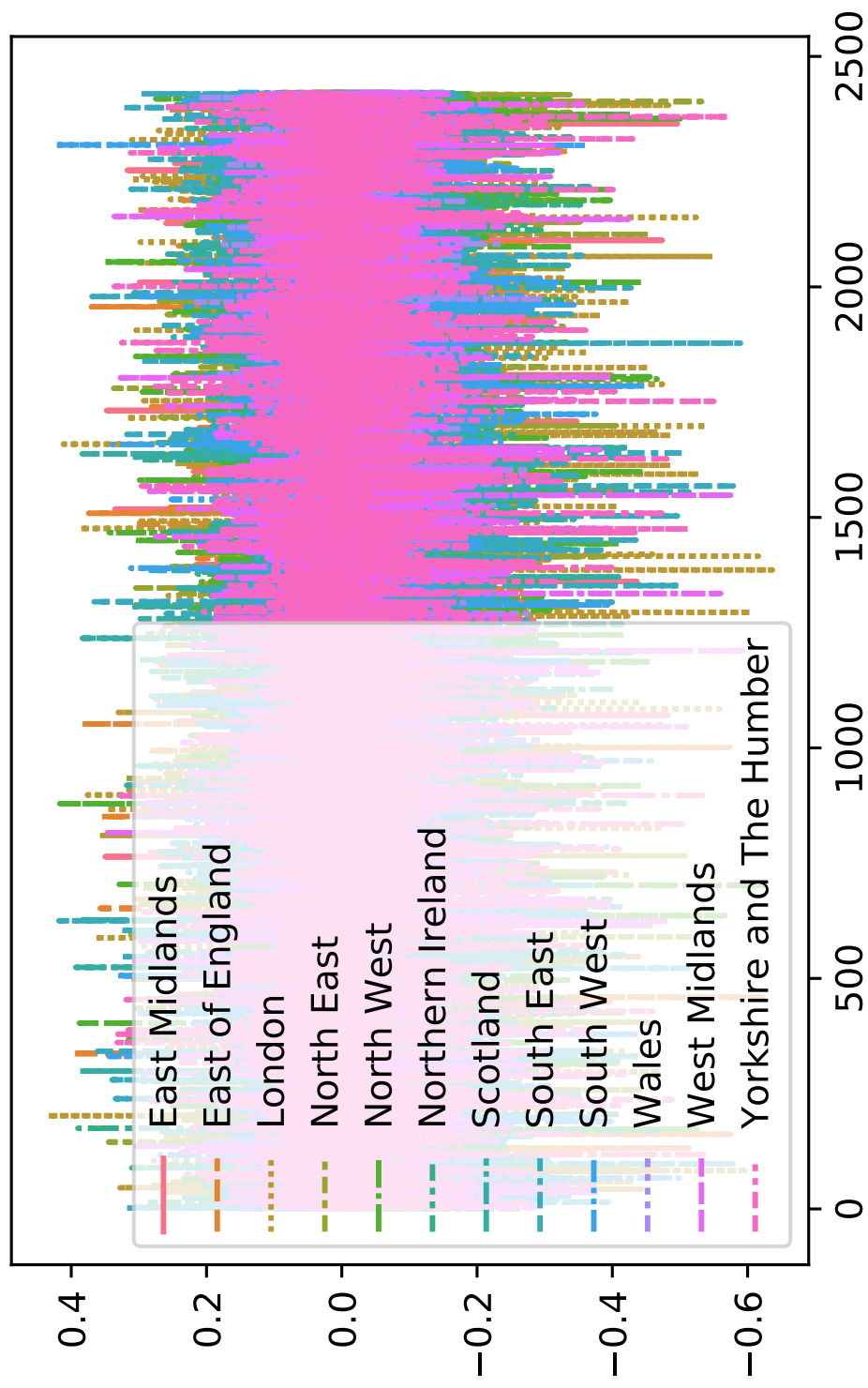
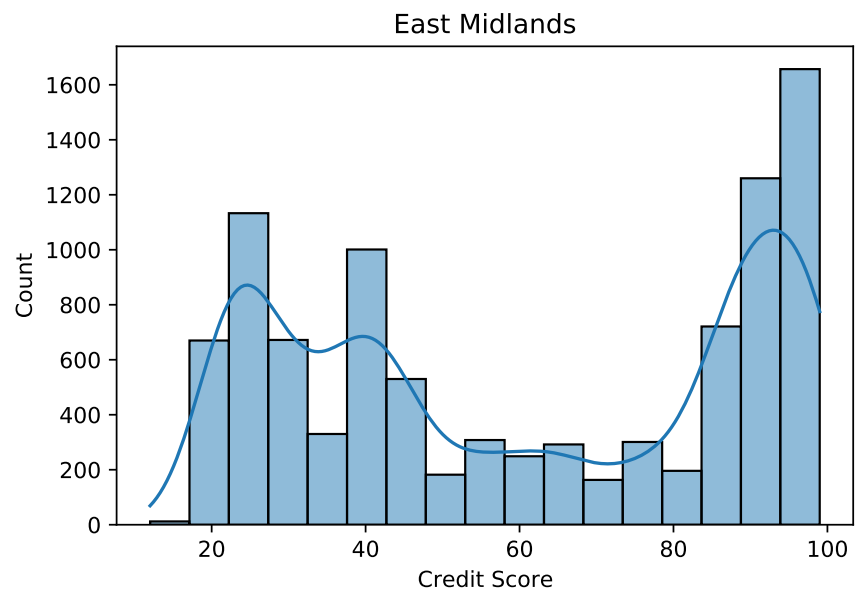


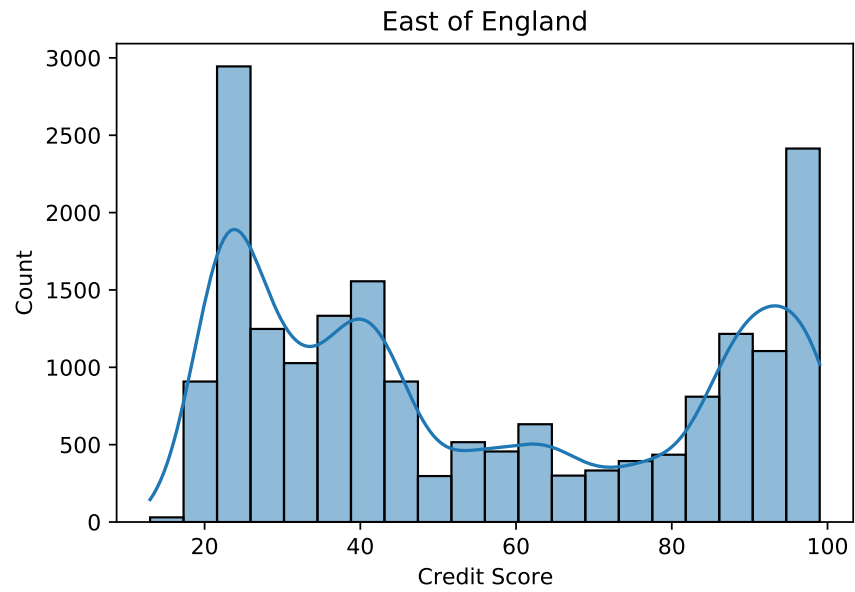
Figure 3. Residuals of estimating credit scores.

Table 12. Estimation performance of the machine learning model for 12 UK regions.

Regions	East Midlands	East of England	London	North East
R^2	0.782121	0.793774	0.750281	0.740741
Regions	North West	Northern Ireland	Scotland	South East
R^2	0.762162	0.714179	0.741691	0.779321
Regions	South West	Wales	West Midlands	Yorkshire & H.
R^2	0.770278	0.772699	0.787782	0.763557

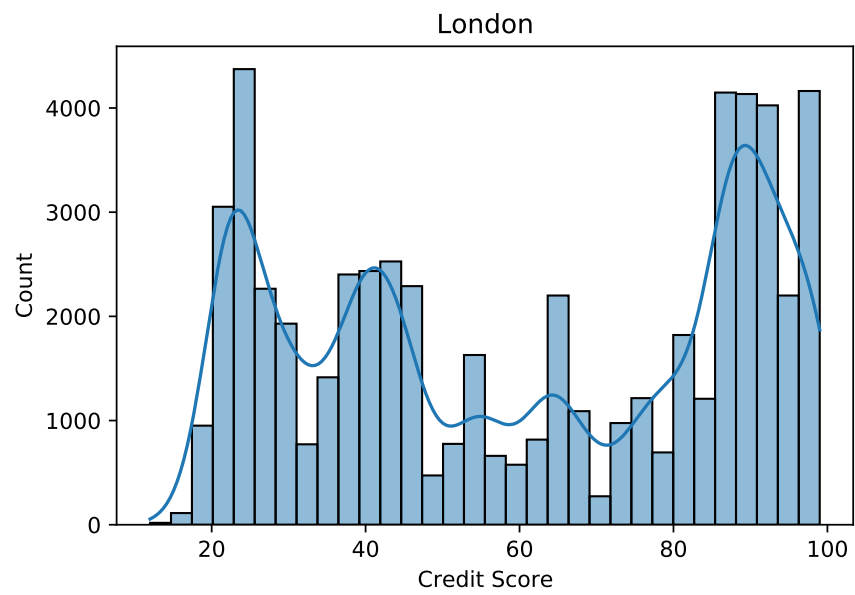


(a)

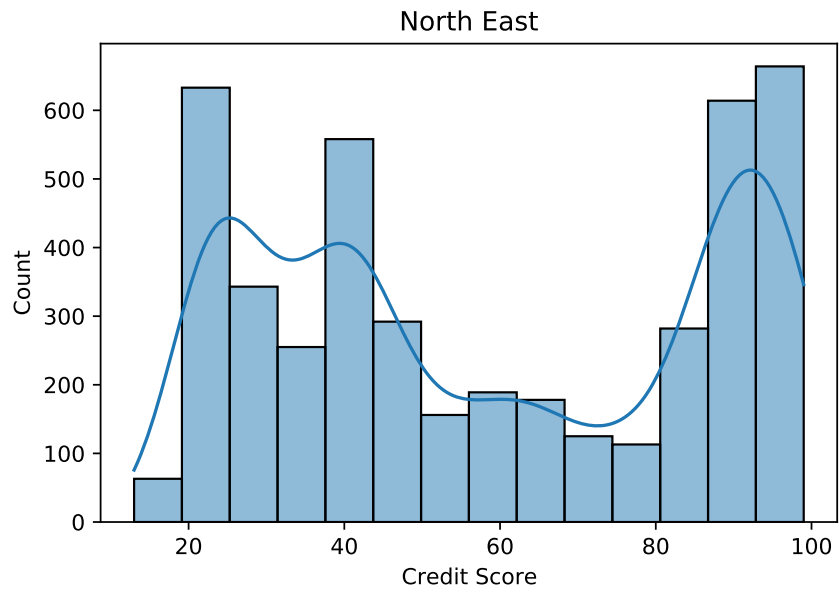


(b)

Figure 4. Distribution of credit scores - group I.

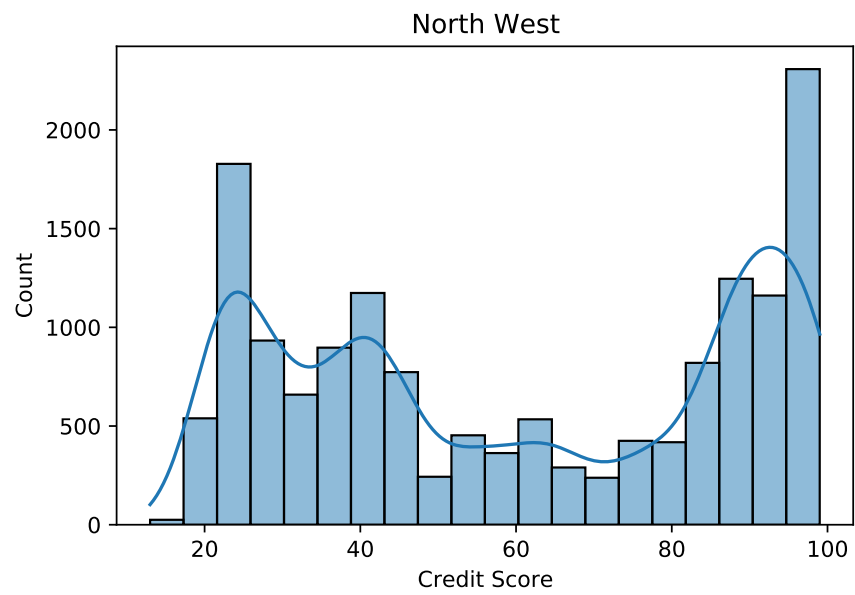


(a)

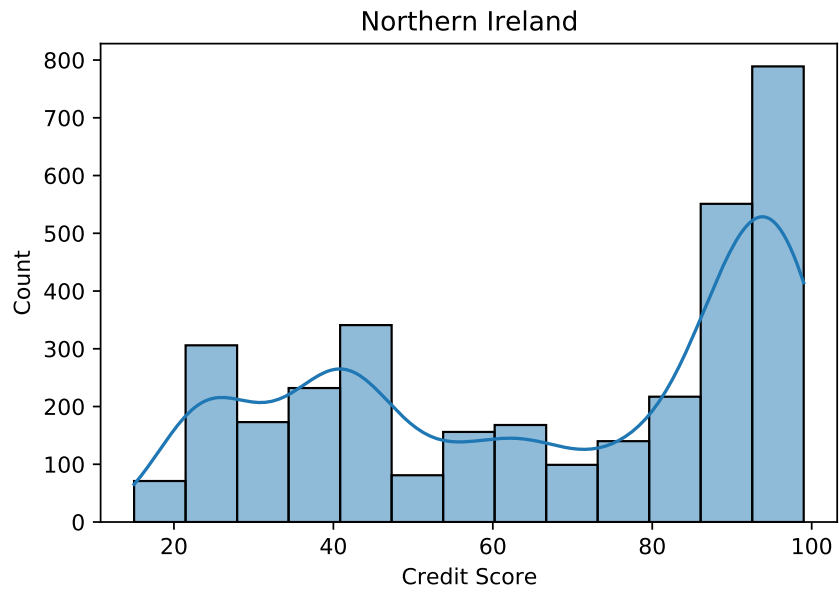


(b)

Figure 5. Distribution of credit scores - group II.

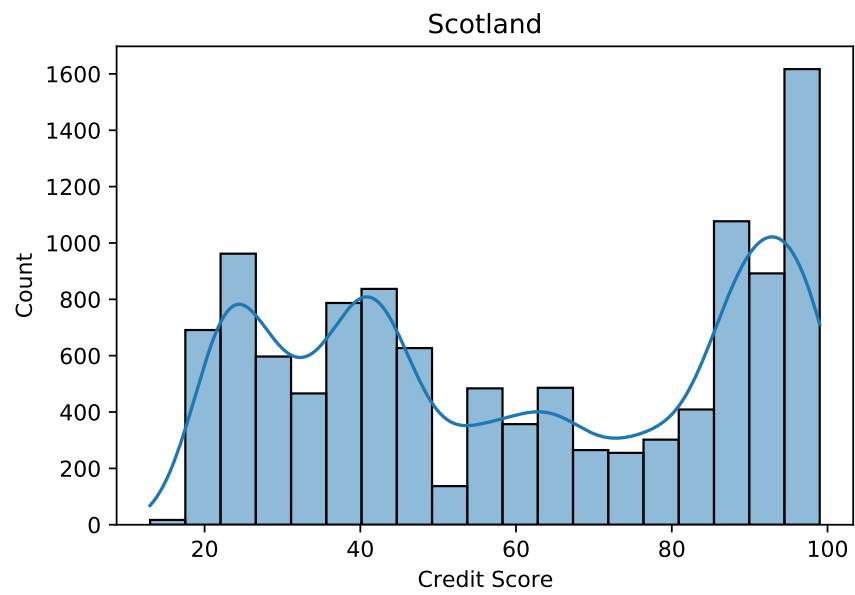


(a)

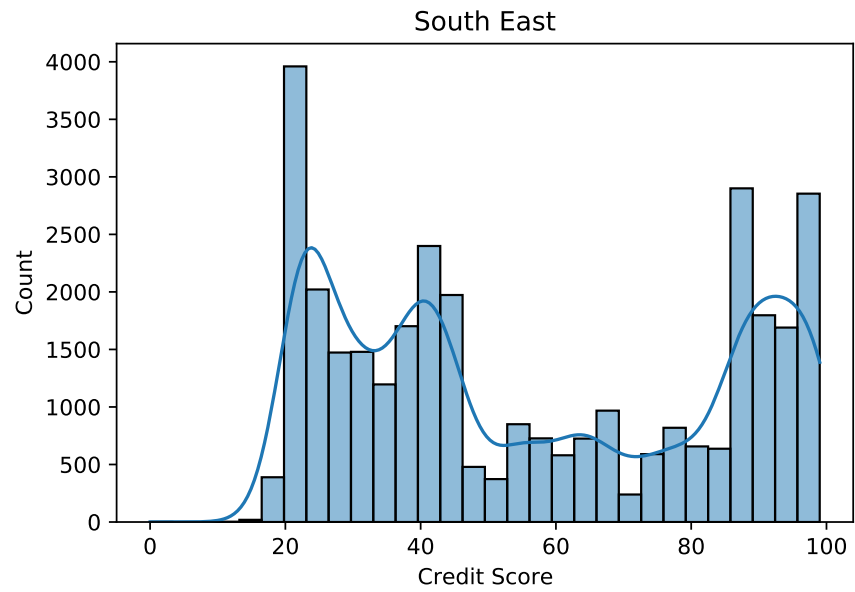


(b)

Figure 6. Distribution of credit scores - group III.

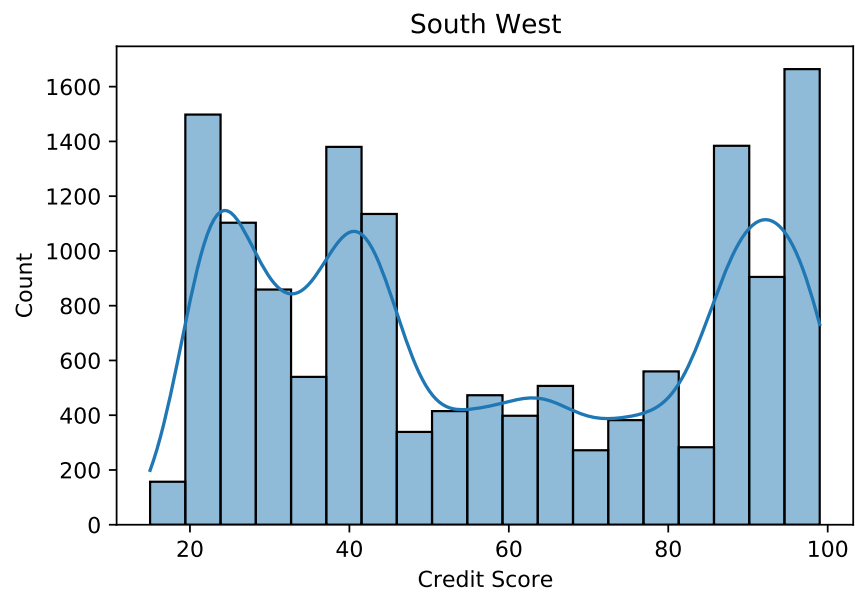


(a)

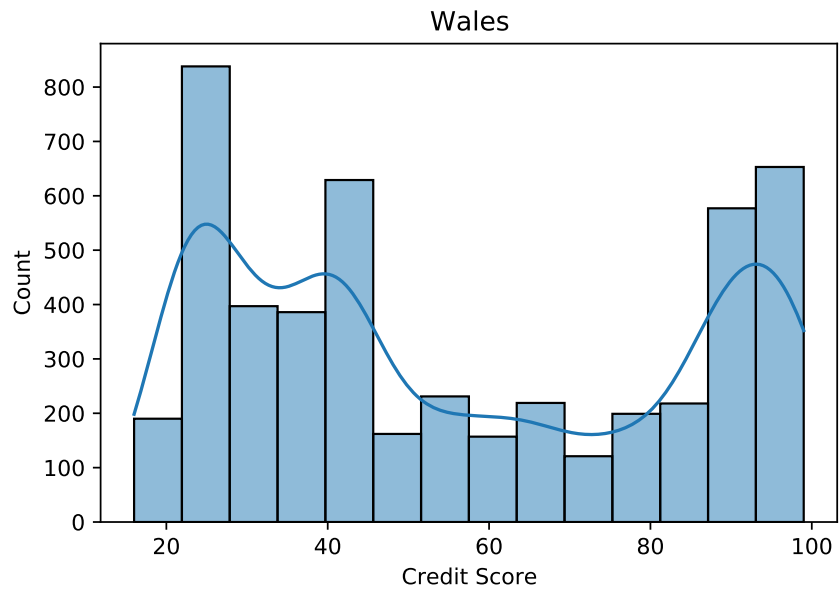


(b)

Figure 7. Distribution of credit scores - group IV.

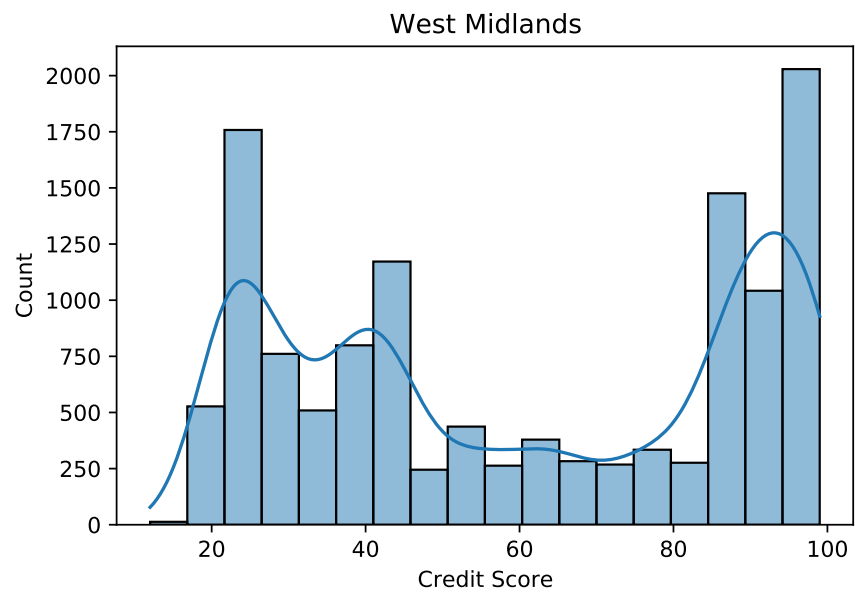


(a)

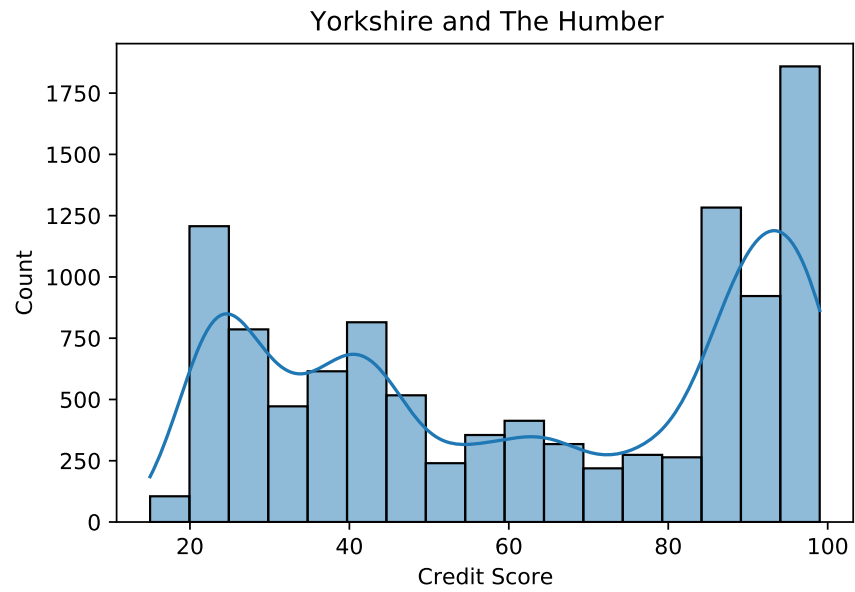


(b)

Figure 8. Distribution of credit scores - group V.

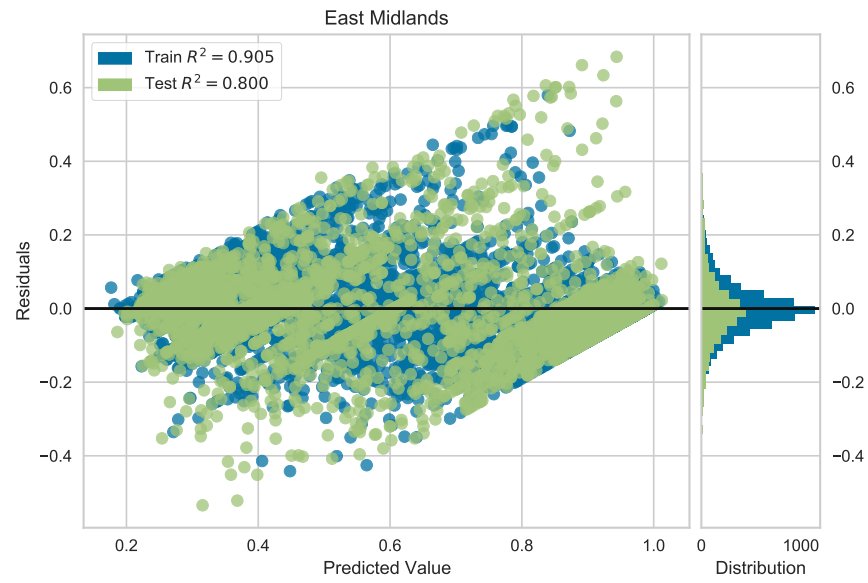


(a)

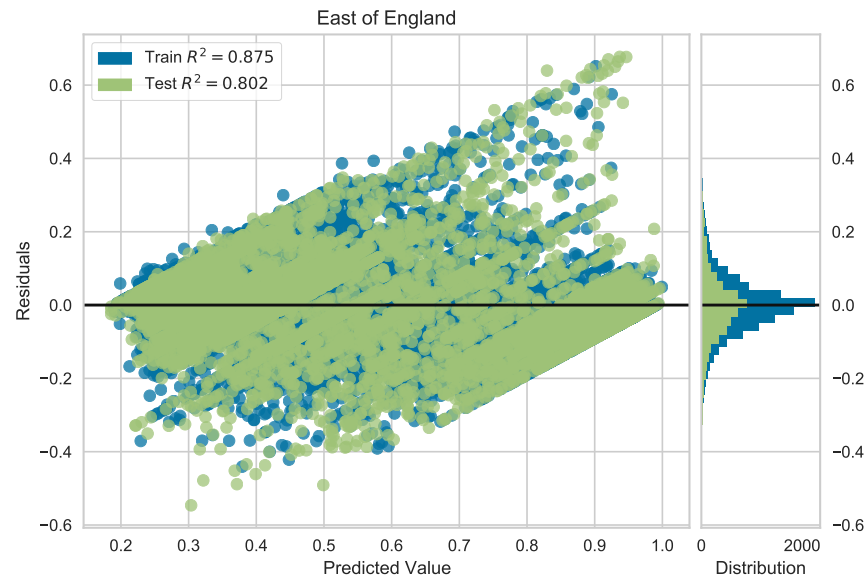


(b)

Figure 9. Distribution of credit scores - group VI.

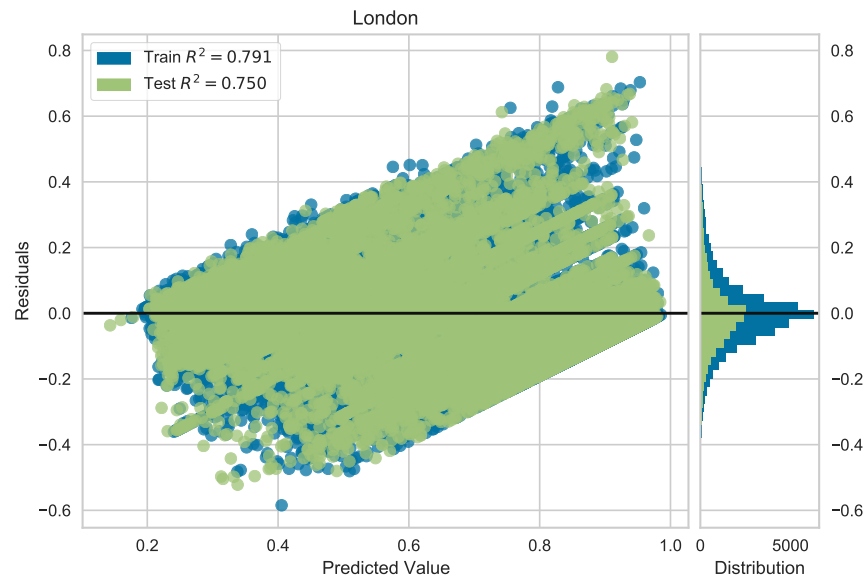


(a)

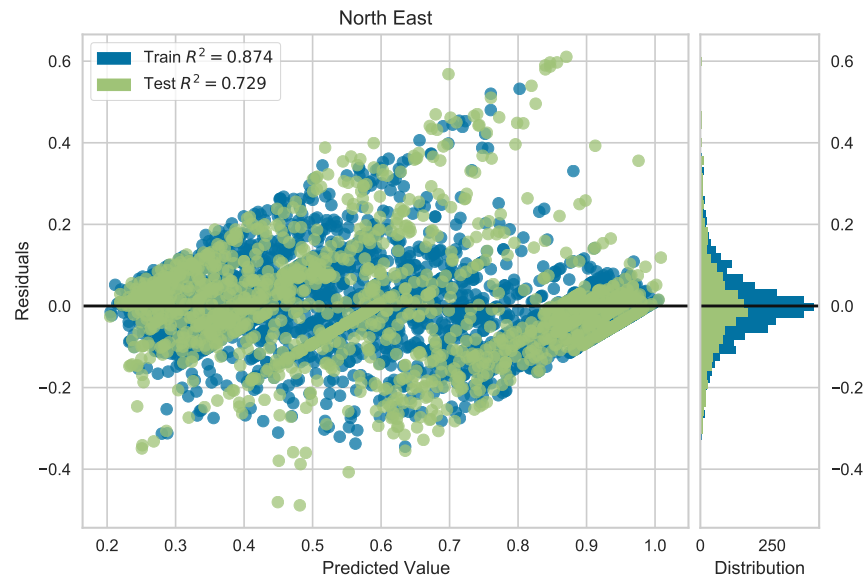


(b)

Figure 10. Residuals of estimating credit scores - group I.

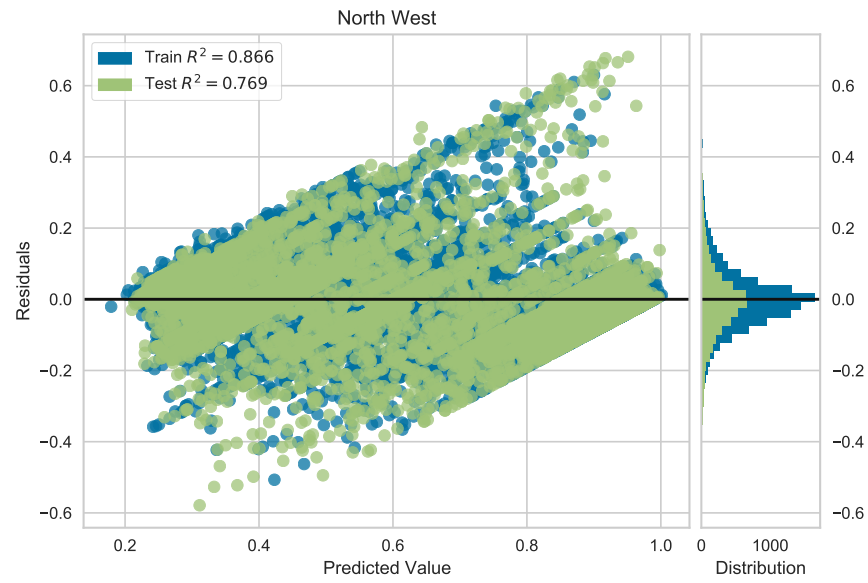


(a)

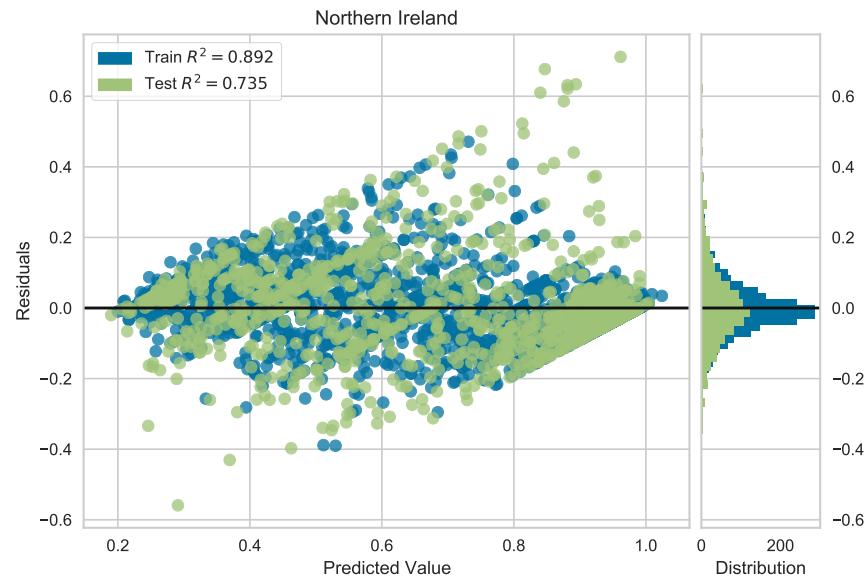


(b)

Figure 11. Residuals of estimating credit scores - group II.

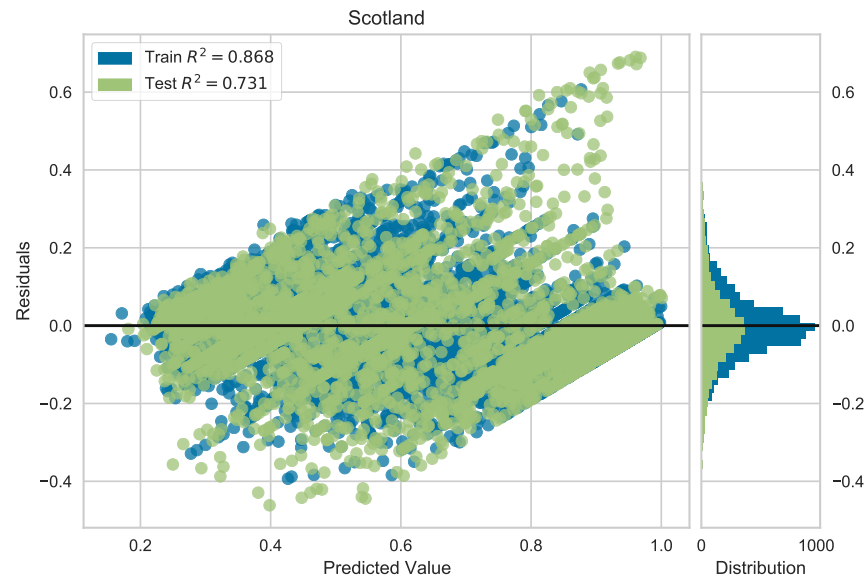


(a)

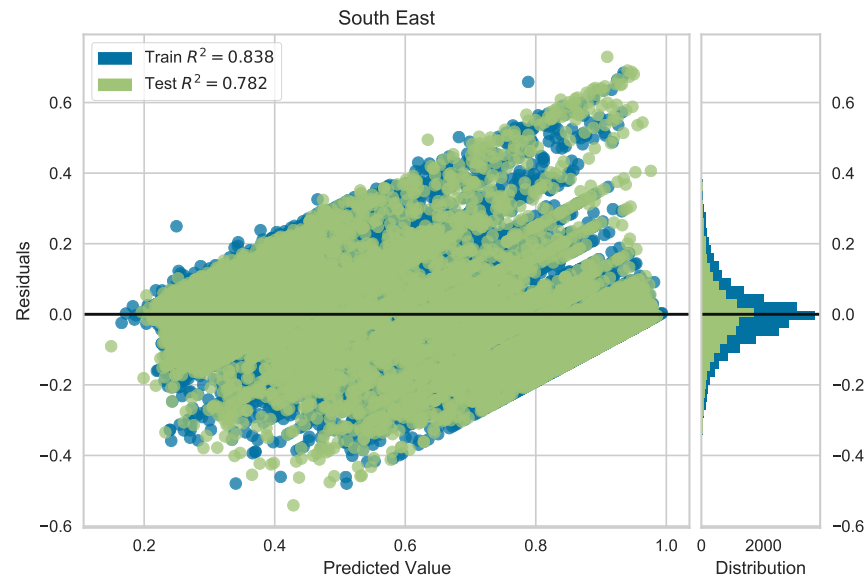


(b)

Figure 12. Residuals of estimating credit scores - group III.

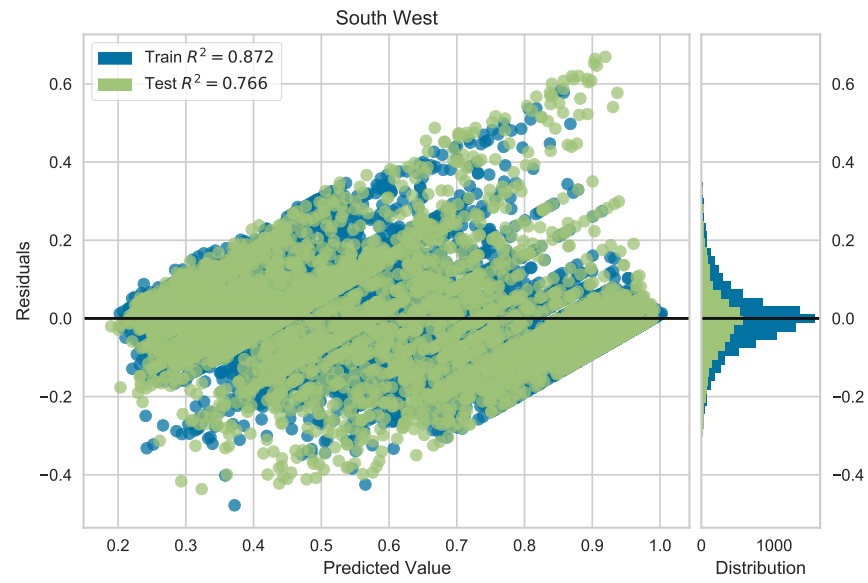


(a)



(b)

Figure 13. Residuals of estimating credit scores - group IV.

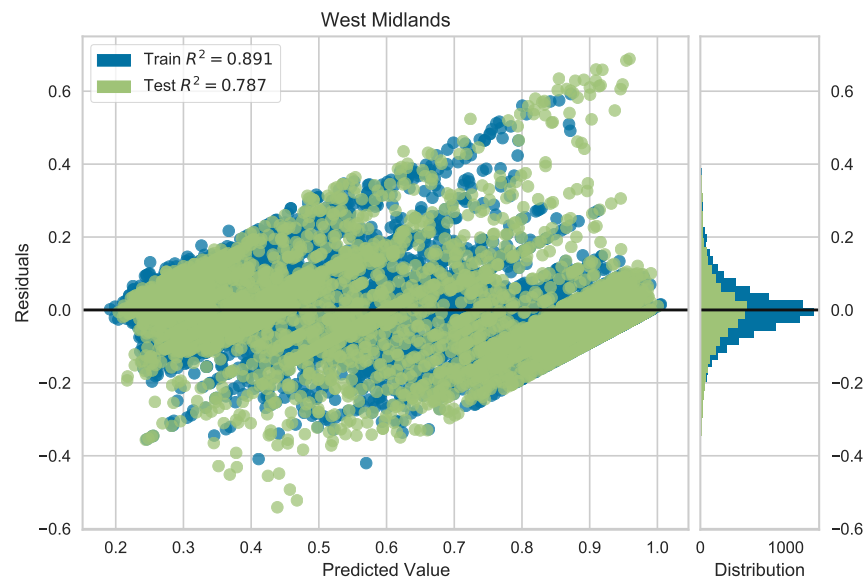


(a)

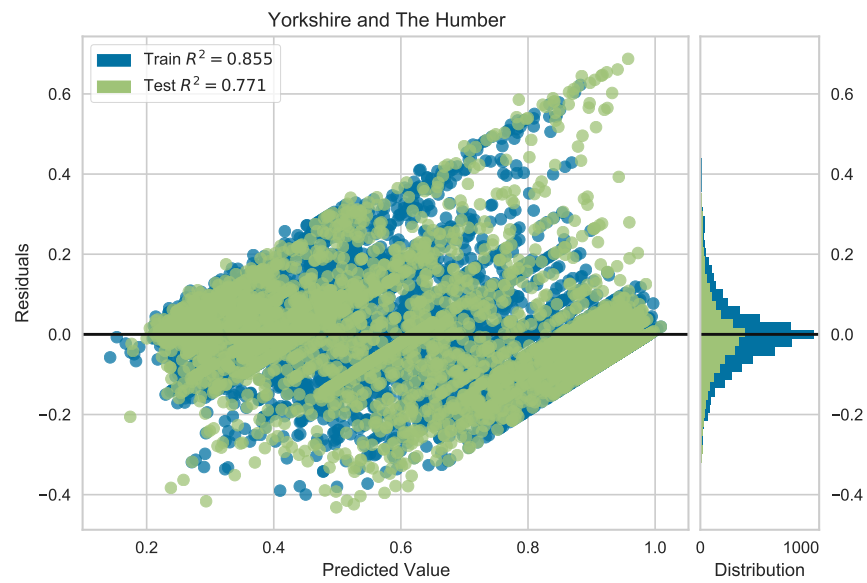


(b)

Figure 14. Residuals of estimating credit scores - group V.



(a)



(b)

Figure 15. Residuals of estimating credit scores - group VI.

7.2 A combination of network analysis, unsupervised ML and supervised ML approaches

We then combine network analysis, supervised ML and unsupervised ML in predicting a firm’s post-COVID credit status. Specifically, we use network analysis and unsupervised ML techniques to uncover the firms with similar credit profiles and then use supervised ML – regression models to predict a firm’s credit score based on the credit status of the firm per se and that of its peers with similar credit profiles. We focus on the firms that have credit information available with regard to 32 credit indicators (see Appendix 1 for the specific variables) in 2019, 2020 and 2021. Our final sample includes 95,776 firms across 88 two-digit SIC (denoted as SIC-2) sectors.

7.2.1 Network analysis

In this section, we explain the method to construct a credit-profile-similarity network. The similarity score between firms is measured based on 32 indicators and is calculated by taking the following steps. First, for each sector, we construct a “feature” matrix where its rows represent the different firms within a sector and the columns represent the firms’ features (i.e., credit profile indicators). Second, we standardize each feature using the min-max scaler, converting the feature into a value ranging from 0 to 1. Feature scaling is an essential step as the feature values are in different ranges and standardization ensures all values are measured and process on the same scale. We use min-max scaler for avoiding having negative feature values which may result in negative similarity scores. Third, we convert the standardized “feature” matrix into a “similarity” matrix by calculating the cosine similarity between all possible firm pairs. Cosine similarity – a widely-used distance measure in

network analysis – is calculated by the following equation:

$$\text{cosine similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

In doing so we obtain the pair-wise similarity score for all possible firm pairs within each sector. On average, the similarity score is 0.82, which is considerably high, causing difficulties for evaluating whether two firms are truly similar. Hence, we use a simulation approach to convert the continuous similarity score into a binary value (1 if two firms are similar in credit profiles and 0 if not).

The simulation approach is taken by the following steps. First, we construct two pseudo firms whose features are randomly selected from the actual value of firms within a certain sector. Second, we calculate the cosine similarity between the two pseudo firms. Third, by repeating this process 1,000 times we obtain 1,000 similarity scores. Fourth, for each sector, we calculate a threshold, which is the 99th percentile of the 1,000 similarity scores obtained from the pseudo firm sample. Then two firms are deemed as truly similar pairs if their actual similarity score is higher than the threshold. There are 290,426,028 potential firm pairs in total. Among them, 17,244,000 are truly similar pairs. As such, the percentage of truly similar pairs is 5.94%.

7.2.2 Unsupervised ML

We also use an alternative approach – K-means cluster analysis (hereafter, K-means), a commonly used unsupervised ML approach – to identify firms with similar credit profiles. Different from the network approach introduced above, K-means does not assess the connectivity (e.g., similarity score) between firms. Instead, it groups similar firms into so-called “clusters”. To ap-

ply K-means, we use the same standardized “feature” matrix as introduced in the prior section, then we use the K-means algorithm provided by the Python package, scikit-learn to conduct the analysis. We use the elbow method to determine the optimal number of clusters for each sector.

Figure 16 presents a visualizable demonstration of a small network and the clustering results based on K-means (SIC-2: 05).

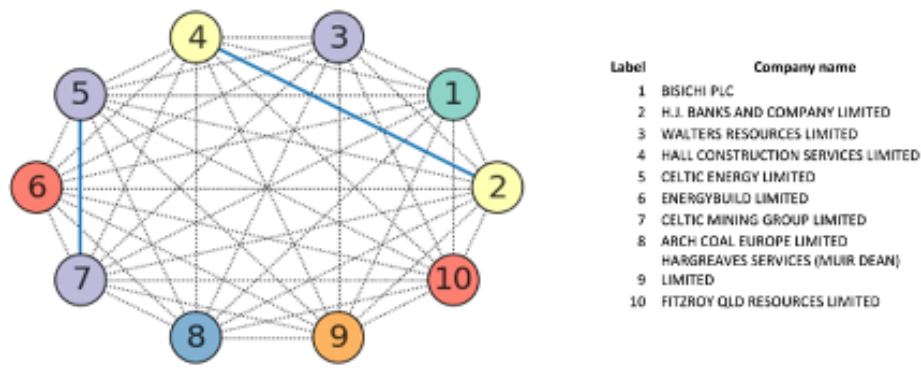


Figure 16. A credit-profile similarity network (SIC-2: 05)

The graph presents a small credit profile similarity network. The sector (SIC-2: 05) contains 10 firms, labelled from 1 to 10 in the nodes. The dot lines represent the potential links between firms. And the blue lines represent “true” linkages between firms (i.e., the firms are truly similar). Nodes in the same color belong to the same clusters identified by the K-means approach.

As shown in Figure 16, Sector 05 has 10 firms in total. Among all 45 potential pairs between the firms, only 2 of them are quantified as true similar pairs. Using K-means, the firms are classified into 6 different clusters.

Figure 17 demonstrates a larger network that is formulated by firms in Sector 01. There are 612 firms classified into 23 different clusters.

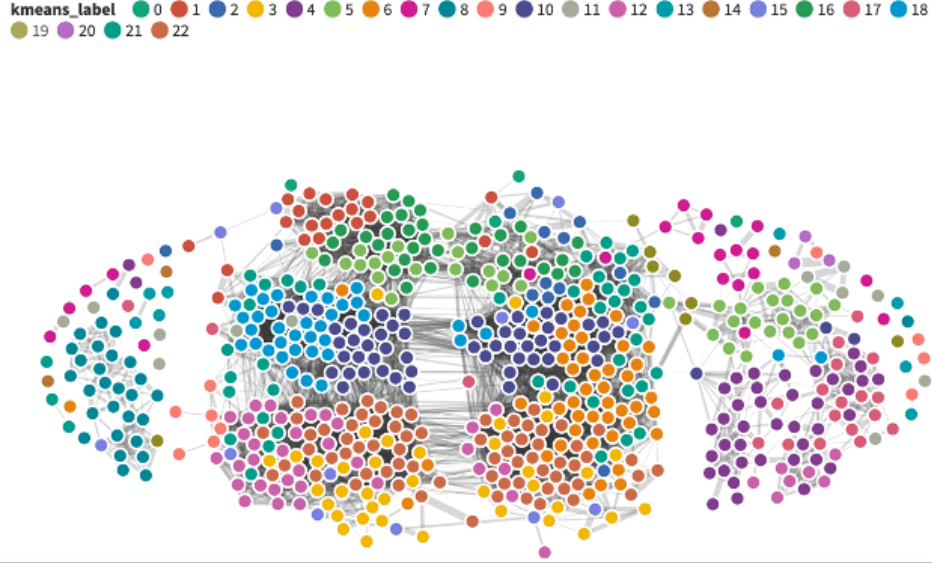


Figure 17. A credit-profile similarity network (SIC-2: 01)

The graph presents a relatively large credit profile similarity network. The sector (SIC-2: 01) contains 612 firms, which are categorized into 23 clusters labelled in different colors.

7.2.3 Network-based credit score and K-means-based credit score

Based on the network connections and K-means clusters, we then calculate the network-based and K-means-based credit scores which are used to predict a firm's post-COVID credit scores. A firm's network-based (K-means-based) credit score is the average pre-COVID credit scores of its network- (K-means-) related peers. In the following, we test empirically the approach introduced above using ANOVA analysis, OLS regression analysis and ML regression analysis.

7.2.4 ANOVA analysis

First, we test whether there are significant differences in pre-COVID / post-COVID credit scores among different K-means clusters. To this end, we apply

ANOVA analysis which is commonly used in testing intra-group differences.
Table 13 demonstrates the results.

Table 13. ANOVA analysis for different sectors

SIC-2	pre-COVID credit score	post-COVID credit score	SIC-2	pre-COVID credit score	post-COVID credit score
01	16.152*** (0.000)	10.999*** (0.000)	50	1.742* (0.040)	1.622+ (0.060)
02	4.656*** (0.000)	4.407*** (0.000)	51	5.013*** (0.000)	1.317 (0.163)
03	3.206** (0.003)	2.075* (0.048)	52	20.625*** (0.000)	11.402*** (0.000)
05	0.646 (0.659)	79.875* (0.012)	53	21.224*** (0.000)	19.236*** (0.000)
06	2.070* (0.030)	0.718 (0.731)	55	17.566*** (0.000)	6.249*** (0.000)
07	3.528+ (0.066)	0.495 (0.859)	56	54.738*** (0.000)	32.969*** (0.000)
08	1.636+ (0.076)	1.651+ (0.075)	58	11.106*** (0.000)	7.358*** (0.000)
09	3.518*** (0.000)	2.127** (0.005)	59	18.606*** (0.000)	14.537*** (0.000)
10	23.810*** (0.000)	7.567*** (0.000)	60	2.911*** (0.000)	1.903* (0.023)
11	4.505*** (0.000)	1.252 (0.250)	61	17.721*** (0.000)	8.588*** (0.000)
12	0.919 (0.507)	0.941 (0.499)	62	144.648*** (0.000)	97.153*** (0.000)
13	612.193*** (0.000)	5.771*** (0.000)	63	18.125*** (0.000)	9.720*** (0.000)
14	6.884*** (0.000)	2.762*** (0.001)	64	15.997*** (0.000)	9.083*** (0.000)

15	7.915*** (0.000)	3.610* (0.010)	65	5.260*** (0.000)	1.932** (0.006)
16	23.174*** (0.000)	13.928*** (0.000)	66	9.793*** (0.000)	5.439*** (0.000)
17	455.184*** (0.000)	120.088*** (0.000)	68	24.600*** (0.000)	13.782*** (0.000)
18	21.575*** (0.000)	8.894*** (0.000)	69	69.361*** (0.000)	57.430*** (0.000)
19	1.188 (0.363)	0.981 (0.498)	70	157.714*** (0.000)	97.754*** (0.000)
20	4.795*** (0.000)	3.021*** (0.000)	71	54.757*** (0.000)	34.819*** (0.000)
21	4.459*** (0.000)	1.690+ (0.059)	72	6.038*** (0.000)	5.720*** (0.000)
22	14.018*** (0.000)	6.946*** (0.000)	73	13.785*** (0.000)	10.194*** (0.000)
23	4.024*** (0.000)	2.429** (0.001)	74	64.090*** (0.000)	31.430*** (0.000)
24	2.238** (0.001)	2.063** (0.003)	75	11.605*** (0.000)	6.864*** (0.000)
25	27.464*** (0.000)	13.269*** (0.000)	77	9.085*** (0.000)	6.248*** (0.000)
26	14.159*** (0.000)	7.951*** (0.000)	78	27.099*** (0.000)	14.490*** (0.000)
27	6.025*** (0.000)	2.656*** (0.000)	79	12.833*** (0.000)	3.717*** (0.000)
28	12.131*** (0.000)	4.358*** (0.000)	80	12.504*** (0.000)	9.745*** (0.000)
29	23.418*** (0.000)	9.444*** (0.000)	81	25.537*** (0.000)	18.409*** (0.000)

30	5.646*** (0.000)	3.466*** (0.000)	82	70.913*** (0.000)	38.668*** (0.000)
31	33.881*** (0.000)	11.093*** (0.000)	84	4.137*** (0.000)	4.901*** (0.000)
32	51.347*** (0.000)	24.444*** (0.000)	85	159.869*** (0.000)	103.768*** (0.000)
33	34.703*** (0.000)	19.077*** (0.000)	86	89.871*** (0.000)	68.420*** (0.000)
35	7.450*** (0.000)	5.417*** (0.000)	87	20.704*** (0.000)	19.083*** (0.000)
36	2.293* (0.017)	1.700+ (0.091)	88	38.427*** (0.000)	31.958*** (0.000)
37	1.068 (0.506)	1.835 (0.261)	90	30.428*** (0.000)	17.381*** (0.000)
38	11.345*** (0.000)	9.810*** (0.000)	91	8.782*** (0.000)	4.583*** (0.000)
39	3.175*** (0.000)	2.425** (0.007)	92	2.833*** (0.000)	1.836* (0.032)
41	78.550*** (0.000)	54.029*** (0.000)	93	26.748*** (0.000)	17.671*** (0.000)
42	39.167*** (0.000)	26.558*** (0.000)	94	28.195*** (0.000)	18.442*** (0.000)
43	152.574*** (0.000)	121.920*** (0.000)	95	12.744*** (0.000)	7.114*** (0.000)
45	49.717*** (0.000)	26.059*** (0.000)	96	37.782*** (0.000)	26.700*** (0.000)
46	153.205*** (0.000)	78.527*** (0.000)	97	8.356* (0.032)	42.897** (0.002)
47	79.767*** (0.000)	51.566*** (0.000)	98	3.699*** (0.000)	4.066*** (0.000)

49	67.343*** (0.000)	51.890*** (0.000)	99	0.882 (0.600)	0.993 (0.499)
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The table presents the results of ANOVA analysis. We run separately the ANOVA for each SIC-2 sector on pre-COVID credit score and post-COVID credit score. F-values are presented in the table. Significance levels are indicated by +, *, **, *** for 10%, 5%, 1% and 0.1% respectively.

As shown in Table 13, there are significant differences between different K-means clusters in the majority of 88 sectors tested in this research. Notably, the significance of pre-COVID credit score is higher than that of post-COVID credit score. This is not a surprise since the K-means clusters are derived based on pre-COVID credit indicators. The results suggest that the clusters may help to explain pre-COVID / post-COVID credit scores.

7.2.5 OLS regression analysis

We then use OLS regressions in testing the predictivity of the K-means-based and network-based credit score on a firm's post-COVID credit score. The dependent variable is a firm's post-COVID credit score. The independent variables are its pre-COVID and the K-means-based and network-based credit scores, and we control a number of other variables that may affect post-COVID credit scores, including the firm's pre-COVID credit score and a series of credit-related indicators. Table 14 presents the results.

Table 14. Regression analysis predicting post-COVID credit score

	Model 1	Model 2	Model 3
Intercept	32.450*** (<0.001)	9.501 (0.177)	-0.274 (0.969)

K-means-based credit score	0.286*** (<0.001)		
Network-based credit score		0.306*** (<0.001)	
Pre-COVID credit score	0.615*** (<0.001)	0.543*** (<0.001)	0.550*** (<0.001)
Self-isolation	-1.155 (0.874)	-7.307 (0.308)	-12.448+ (0.083)
Overseas turnover%	1.540*** (<0.001)	-0.803* (0.037)	-1.054** (0.007)
Gearing	-0.002+ (0.073)	-0.000 (0.814)	0.001 (0.623)
Asset turnover	0.017*** (<0.001)	0.017*** (<0.001)	0.016*** (<0.001)
Creditors payment days	0.018*** (<0.001)	0.012*** (<0.001)	0.011*** (<0.001)
Debtor collection days	10.905*** (<0.001)	6.389*** (<0.001)	6.647*** (<0.001)
Exceptional items	184.246** (0.002)	205.700*** (0.001)	217.379*** (<0.001)
Balance sheet minorities	963.940*** (<0.001)	522.761* (0.015)	443.798* (0.039)
Regional GDHI per head	-0.000 (0.571)	-0.000 (0.636)	-0.000 (0.818)
COVID information	4.558 (0.673)	9.386 (0.379)	16.457 (0.124)
Affordability	-15.193 (0.293)	-13.134 (0.357)	-16.277 (0.255)
Current liabilities%	-6.799*** (<0.001)	-4.485*** (<0.001)	-3.991*** (<0.001)

Geographical diversification	1.168*** (<0.001)	0.587*** (<0.001)	0.805*** (<0.001)
Firm age	0.000*** (<0.001)	0.000*** (<0.001)	0.000*** (<0.001)
Administration expenses%	-1.203*** (<0.001)	-0.497*** (0.001)	-0.542*** (<0.001)
Debtors collection	-0.002 (0.298)	-0.000 (0.936)	0.001 (0.662)
SME	0.496* (0.015)	2.451*** (<0.001)	3.436*** (<0.001)
Firm independence	1.348*** (<0.001)	1.453*** (<0.001)	1.650*** (<0.001)
EBIT%	0.323* (0.04)	0.721*** (<0.001)	0.646*** (<0.001)
Tangible assets%	0.058 (0.764)	0.537** (0.005)	0.589** (0.002)
Pessimism	-10.850 (0.315)	-8.054 (0.449)	-2.682 (0.801)
ROA	-0.545*** (<0.001)	0.380*** (<0.001)	0.079 (0.414)
Business diversification	-0.079 (0.757)	-0.107 (0.672)	-0.059 (0.816)
Liquidity ratio	-0.041*** (<0.001)	-0.020*** (<0.001)	-0.020*** (<0.001)
Severe anxiety	1.496 (0.834)	4.045 (0.566)	9.837 (0.163)
No. of observations	78585	78553	78547
Adj. R-square	0.453	0.467	0.466

This table presents the OLS regression analysis. The dependent variable is post-COVID credit score. The independent variables are K-means-based score and Network-based credit score. K-means-based (Network-based) credit score is the average credit score of a firm’s peers identified by the K-means cluster analysis (Network analysis). We control for a number of credit-related variables including pre-COVID credit score, financial variables, and COVID-related indicators. Significance levels are indicated by +, *, **, *** for 10%, 5%, 1% and 0.1% respectively.

As shown in Table 14, both K-means-based and network-based credit scores demonstrate significant predictability on post-COVID credit scores, even after controlling for a number of related variables. Specifically, the coefficient of K-mean-based credit score is 0.286 ($p < .001$), and the coefficient of network-based credit score is 0.306 ($p < .001$). Notably, there is an increase in the adjusted R-squares after adding both variables into the regression. Model 2, after adding K-means-based credit score, has a 1.4% increase compared to Model 1. And Model 3 has an increase of 1.3% on adjusted R-squares. The results support the ANOVA analysis, suggesting that both network-based and unsupervised ML classifications have explanations for a firm’s post-COVID credit score.

7.2.6 ML regression models

Lastly, we construct ML regression models, using the pre-COVID credit scores, credit-related indicators, and K-means-based/network-based credit scores to predict a firm’s post-COVID credit score. We use a number of ML regression models including XGB, LGBM, KNN, decision tree, Ada boost, bagging, extra trees and Gradient boosting. The R-squares of using different regression models are demonstrated in Figure 18.

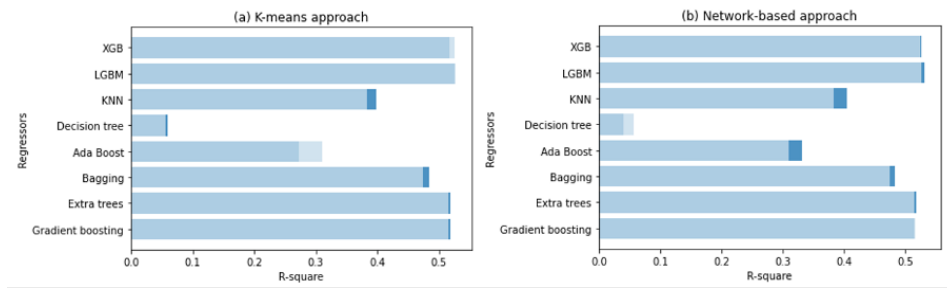


Figure 18. R-squares of machine learning regression models

The graph presents the R-squares of different ML regression models. The bars in the graphs are labelled in three colors. The light blue areas represent the extent to which by adding K-means-based (Network-based) credit score the models achieve lower R-squares. The dark blue areas represent the extent to which by adding K-means-based (Network-based) credit score the models achieve higher R-squares.

As it is shown in Figure 18, the regression models achieved moderate predictability on post-COVID credit scores. Among all regression models, the extra-trees model appears to be most effective, having an R-square around 55%. By adding K-means/Network-based credit scores into the regression models, most regression models have a marginal increase, particularly KNN and Bagging model.

8 Conclusions

This study aims to enhance the prediction models of post-covid solvency of UK firms (with a particular focus on SMEs). We measure post-covid solvency using credit scores and the value of county court judgements (total and unsatisfied) after 1st April 2020. Our approach is to combine as many variables as possible into few firm-level, regional, and sectoral indices. We utilize Principal Component Analysis to construct firm-level financial, non-financial, and overall indices and a regional social covid impact index. We also utilize stock market reactions around the first UK national lockdown announcement on 23rd March 2020 to construct a sectoral covid exposure index. Our OLS regression results show that our indices have statistically and economically significant predictive power of post-covid solvency in UK firms and SMEs.

Venues for further research can include: 1) using Experian data (rather than FAME data) to construct firm-level financial, non-financial, and overall indices, 2) using CBI, ONS, and GOV data to construct more regional and sectoral indices, 3) using local authority data (rather than NUTS1 data) to construct regional indices, 4) Using more measures of post-covid solvency of UK firms and SMEs such as the probability of default, Experian's credit ratings and risk indices, lender-borrower relationships, and covenants of post-covid credit facilities, 5) validating our indices by using longer time-series (for example, 2010-2021 rather than 2019-2021), and 6) using more advanced data reduction techniques such as Autoencoder¹⁶ (for example, see [Gu et al. \(2021\)](#))

¹⁶ [Gu, Kelly and Xiu \(2021\)](#), p.430 say "Autoencoders are workhorse dimension reduction models in the field of machine learning. They can be thought of as a nonlinear, neural network counterpart to principal components analysis (PCA). Both autoencoders and PCA are unsupervised methods - they attempt to model the full panel of asset returns using only the returns themselves as inputs."

and [Smiti and Soui \(2020\)](#))).

We use machine learning models to estimate firms’ credit scores based on UK companies from FAME and other data sources. We predict firms’ credit score and examine the residuals and estimation performance. Our sample size is sufficient for the model to achieve stable estimation performance. LightGBM achieves superior performance in credit assessment and is much faster.

We carry out a comprehensive analysis of UK firms’ credit scores in the 12 regions individually. London has a relatively even distribution of credit scores while most other regions show two fat tails. The importance of firm features for the 12 regions are similar, where the likelihood of failure (the firm age) has the most significant negative (positive) effects on credit scores.

We find that all of the residuals of the estimates demonstrate series like white noises. Using classic information criteria to these residuals shows that their autoregressive orders are zeros. Both results indicate that the models estimated for the 12 regions are adequate descriptions of the regional data and there are not further additional structures in the residuals of the estimated model.

We then used cluster analysis and network analysis to uncover firms with similar credit profiles. The ANOVA analysis suggests that firms in different K-means clusters differ in pre-COVID/post-COVID credit scores. Further, the OLS regression analysis suggest that a firm’s post-COVID credit score is determined not only by its pre-COVID credit score but also by that of its similar peers. Based on the evidence, we include the “peer factors” obtained from the K-means approach and the network approach into the ML regression models estimating post-COVID credit score. We document a marginal increase of R-squares in several models suggesting that the unsupervised ML and network analysis enhance the ML performance.

Appendix A List of Financial Variables Excluded from PCA

Note: All balance sheet ratios are denominated by total assets and all income statement ratios are denominated by turnover. Current ratio = Current assets / Current liabilities. Data is collected from FAME.

1. Intangible Assets%
2. Investments (Fixed Assets)%
3. Stock & WIP%
4. Bank & Deposits%
5. Other Current Assets%
6. Investments (Current Assets)%
7. Short-term Loans & Overdrafts%
8. Total Other Current Liabilities%
9. Long-term Liabilities%
10. Long-term Debt%
11. Total Other Long-term Liabilities%
12. Provisions for Other Liabilities%
13. Pension Liabilities%
14. Issued Capital%

15. Gross Profit%
16. Cost of Sales%
17. Other Operating Income Pre-Operating Profit%
18. Pre-tax Income%
19. EBITDA%
20. Current Ratio

Appendix B List of Social Variables Excluded from PCA

Note: Data is collected from the ONS survey "Coronavirus and the social impacts on households in subnational areas in Great Britain: 2020 and 2021": <https://www.ons.gov.uk/releases/...britain2020and2021>.

1. **Working from home:** Proportion of people who have worked from home because of Covid.
2. **Avoiding physical contact:** Proportion of people who have avoided physical contact with others outside own home because of Covid.
3. **Level of worry:** Level of worry about the effect of Covid on personal life (very worried, somewhat worries, neither, somewhat unworried, not worried at all).
4. **Ability to save:** Proportion of people who think they are able to save money in the next twelve months after Covid.

5. **Left home (for any reason):** Proportion of people who have left home (for any reason) after Covid.
6. **Left home for work:** Proportion of people who have left home (for work) after Covid.
7. **Left home to meet people:** Proportion of people who have left home (to meet people) after Covid.
8. **Left home for essentials:** Proportion of people who have left home (for essential purposes) after Covid.
9. **Support for lockdown measures:** Proportion of people who support lockdown measures where they live.

Appendix C List of Credit-related Variables Used in ML/Network Analysis

Variable		Variable	
1	Total Assets	17	Assets Turnover
2	Fixed Assets%	18	Gearing
3	Tangible Assets%	19	Debtors Collection
4	Debtors%	20	Creditors
5	Current Liabilities	21	SME
6	Creditors%	22	Firm Age
7	Balance Sheet Minorities%	23	Number of Employees
8	Shareholders Funds	24	Firm Independence
9	Total Reserves%	25	Business Diversification
10	Turnover	26	Geographical Diversification

11	Overseas Turnover%	27	Severe Anxiety
12	EBIT%	28	Covid Information
13	Administration Expenses%	29	Self-isolation
14	Exceptional Items%	30	Affordability
15	ROA	31	Pessimism
16	Liquidity Ratio	32	Regional GDHI Per Head

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